

1 INTRODUCTION TO ROBUST CONTROL

1.1 Introduction

In this chapter, we turn our attention to the issue of robustness. The design of highly accurate control systems in the presence of significant uncertainty requires the control engineer to seek a robust system.

A robust system exhibits the desired performance despite the presence of significant plant (process) uncertainty.

Control engineering always deals with the analysis and design of control systems for plants which contain physical parameters subject to perturbations. The physical parameters could consist of masses, inertias, spring constants, aerodynamic coefficients etc., that are required in the mathematical description of the dynamics of the plant. In engineering, one frequently encounters situations where the structure of the plant and nominal values of these parameters are known, but the parameters undergo large perturbations as the operating conditions of the control system change. Such problems cannot be treated within the framework of the familiar control theory.

The design of a control system is invariably based on an assumed nominal model of the plant to be controlled. However, the plant model will always be an inaccurate representation of the actual physical system because of

Parameter changes

Unmodelled dynamics

Unmodelled time delays

Changes in equilibrium point (operating point)

Sensor noise

Unpredicted disturbance inputs

After the usual simplifications, such as linearization about an operating point, lumped parameter approximation etc., one ends with up most often with a linear time-invariant system described by a prescribed set of differential equations for the nominal model of the plant. At this point, various established design strategies such as the frequency domain methods of classical control, or the state space methods of optimal control theory can be applied to the nominal model to produce a feedback controller that yields closed-loop stability and an acceptable output time response.

An important and fundamental practical requirement to be satisfied by the controller is invariance of the property of closed-loop stability, under perturbations, from a suitable class, of the nominal plant model. A controller satisfying this requirement is said to be robust with respect to the prescribed class of perturbations. The theory of analysis and design of such controllers is currently an active area of research in control theory. A system subject to perturbations is said to be uncertain.

Two main classes of problems in robust control are robust synthesis and robust analysis. In general, robust synthesis problems are much more difficult.

The specific of the class of perturbations against which robustness is required depends, of course, on the physics and engineering of the particular plant in question. The general theory, however, distinguishes broadly between the following types of perturbation classes:

Structured and unstructured

Linear and nonlinear

Time-invariant and time-varying.

Example 1.1

Consider two mass spring system shown in Fig. 1.1. In this case, the mass perturbations are linear and time-invariant, the spring perturbations may be considered as non-linear and time-

varying. This may reflect the non-linear nature of the true spring.

In this chapter, we will consider the issue of robust stability.

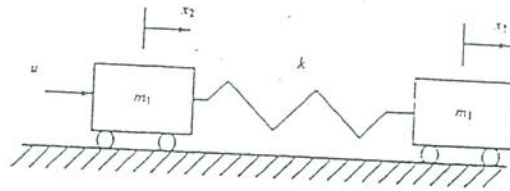


Fig. 4.1: Two mass spring system.

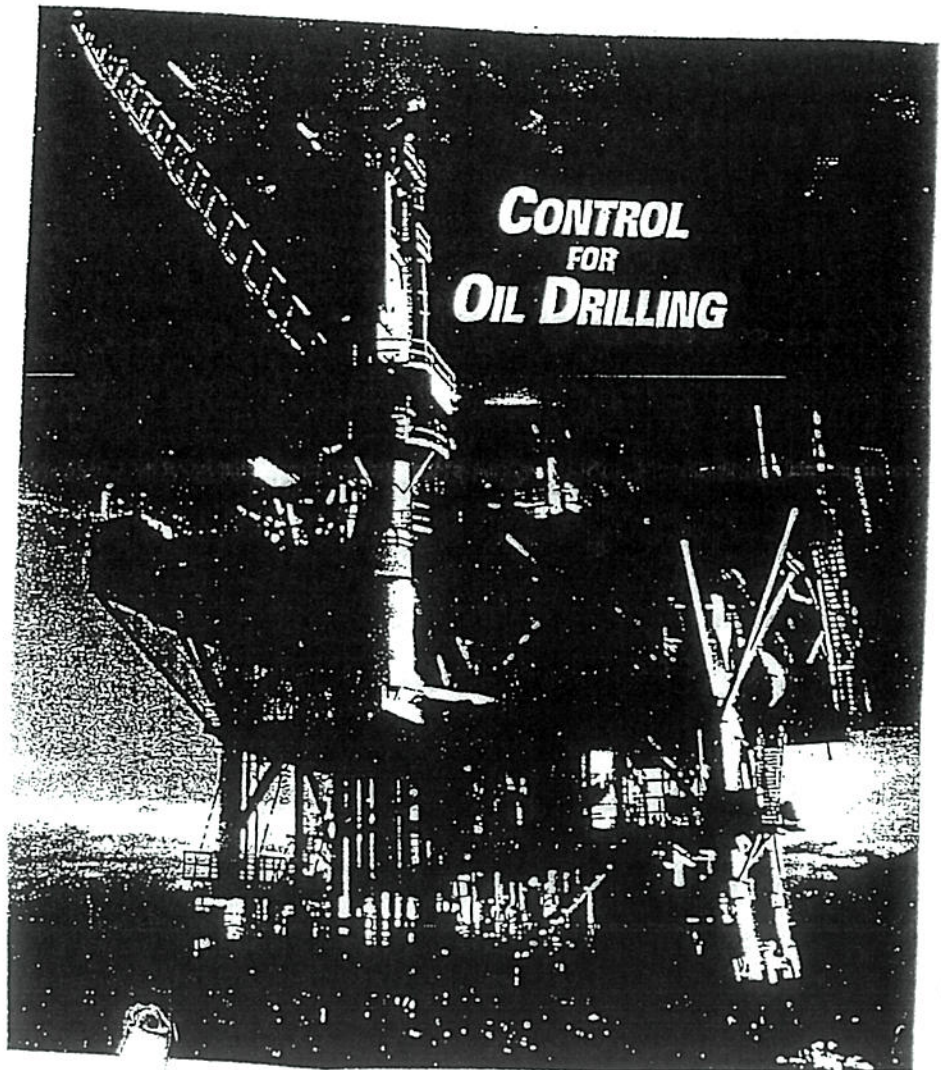


Fig 4.2(a)

1.2 Kharitonov's Theorem

Consider a linear time-invariant continuous-time uncertain system subject to linear time-invariant perturbations. The characteristic polynomial of such a system is a polynomial with coefficients subject to perturbations.

Consider the following polynomial

$$a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0$$

where

$$a_0 \in [m_0, M_0], a_1 \in [m_1, M_1], \dots, a_n \in [m_n, M_n].$$

Such a polynomial is called an interval polynomial.

An interval polynomial is said to be stable if the above polynomial is stable for any coefficients from the above intervals.

The question is how to study stability for an interval polynomial.

Theorem (Kharitonov). Consider the following four polynomials:

$$M_n s^n + m_{n-1} s^{n-1} + m_{n-2} s^{n-2} + M_{n-3} s^{n-3} + M_{n-4} s^{n-4} + \dots,$$

$$M_n s^n + M_{n-1} s^{n-1} + m_{n-2} s^{n-2} + m_{n-3} s^{n-3} + M_{n-4} s^{n-4} + \dots,$$

$$m_n s^n + M_{n-1} s^{n-1} + M_{n-2} s^{n-2} + m_{n-3} s^{n-3} + m_{n-4} s^{n-4} + \dots,$$

$$m_n s^n + m_{n-1} s^{n-1} + M_{n-2} s^{n-2} + M_{n-3} s^{n-3} + m_{n-4} s^{n-4} + \dots.$$

These polynomials are called extreme. Then, the interval polynomial is stable if and only if these four polynomials are stable.

Note that Kharitonov's theorem is not true for discrete-time systems.

A general procedure to check stability of an interval polynomial:

1. Find four extreme polynomials.
2. Check stability of these four polynomials.

Example 1.2

Consider the following interval polynomial

$$s^3 + a_2 s^2 + a_1 s + a_0$$

where

$$a_2 \in [2, 10], a_1 \in [3, 5], a_0 \in [3, 4].$$

Study stability.

Solution This interval polynomials has the following extreme polynomials:

$$s^3 + 2s^2 + 3s + 4,$$

$$s^3 + 10s^2 + 3s + 3,$$

$$s^3 + 10s^2 + 5s + 3,$$

$$s^3 + 2s^2 + 5s + 4.$$

Using Routh's stability criterion, we obtain that all these polynomials are stable. Hence, the interval polynomial is stable.

Kharitonov's theorem deals with the case of linear time-invariant structured perturbations.

6.3 Edge Theorem

Consider now the following uncertain polynomial

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0$$

where

$$[a_0, a_1, a_2, \dots, a_{n-1}, a_n] \in \Omega$$

and $\Omega \subset \mathbb{R}^{n+1}$ is a polytope.

The question is how to check stability of such uncertain polynomials. Kharitonov's theorem gives the answer in the case when Ω is a box. How about the general case?

Edge Theorem. The above interval polynomial is stable if and only if all polynomials with coefficients corresponding to the edges of the polytope Ω are stable.

Example 1.3

Stick-slip oscillations in oil well drillstrings are an example of limit-cycling behavior. In mechanical systems, limit cycles occur due to hysteresis, backlashes between contacting parts, dry friction between sliding parts, nonlinear damping, geometrical imperfections, etc. For the oil well drillstrings considered here (see Fig. 6.2(a) and (b)), dry friction characteristic, often referred to as stick-slip, may cause periods of persistently self-exciting oscillations. The characteristic polynomial of a PD-like control system for suppression of stick-slip oscillations is given by

$$s^3 + a_2 s^2 + a_1 s + 1$$

where $a_1 \in [1, 3]$, $a_2 \in [1, 4]$ and $a_1 + a_2 \geq 3$. Study stability.

Solution Here $[a_1, a_2] \in \Omega$ where Ω is the polytope on the (a_1, a_2) plane shown in Fig. 1.3.

Let us check stability by the use of Edge Theorem.

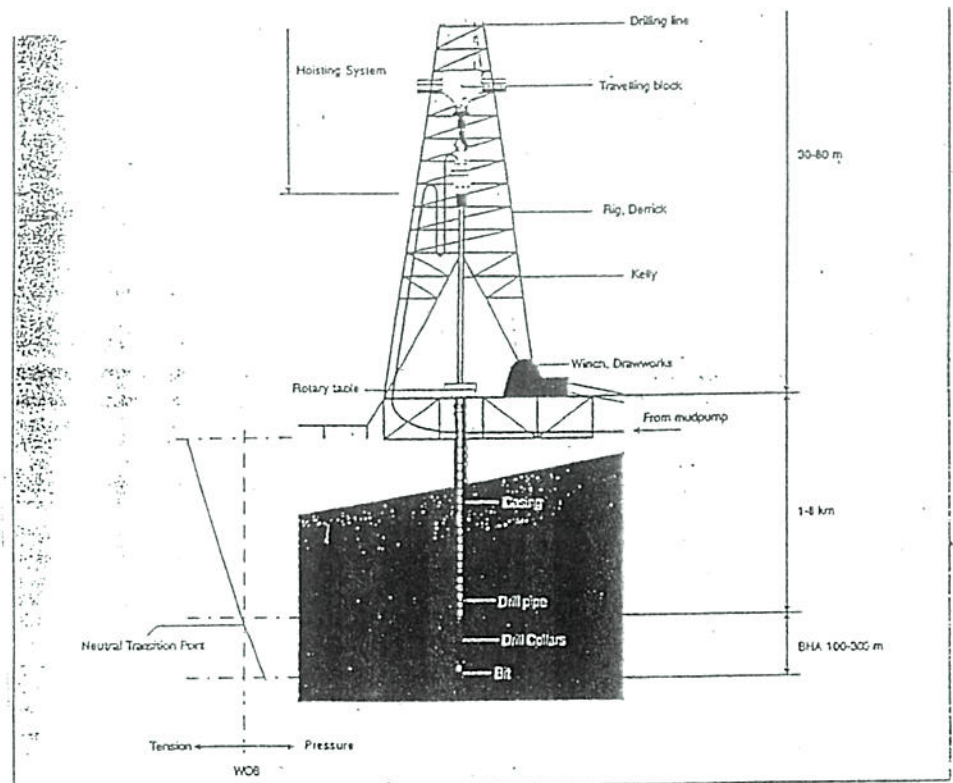


Fig. 1.2(b)

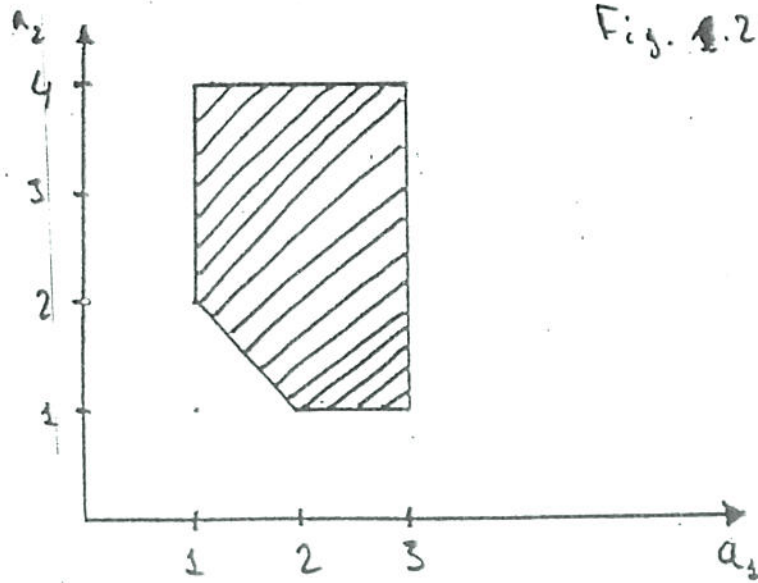
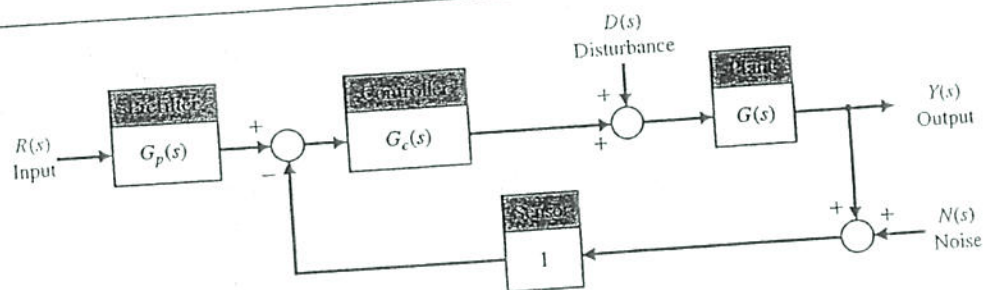


Fig. 1.3. - 2 -

The system structure that incorporates potential system uncertainties is shown in Fig. 12.1. This model includes the sensor noise $N(s)$, the unpredicted disturbance input $D(s)$, and a plant $G(s)$ with potentially unmodeled dynamics or parameter changes. The unmodeled dynamics and parameter changes may be significant or very large, and for these systems, the challenge is to create a design that retains the desired performance.

FIGURE 12.1
Closed-loop
system structural
diagram.



2 CLASSICAL APPROACH TO ROBUST CONTROL DESIGN

In this chapter, we apply methods of classical control (Bode design) to the problem of robust control system design.

This chapter is Chapter 12 of [1].

[1] R.C.Dorf and R.H.Bishop. Modern Control Systems. Addison-Wesley, 8th edition, 1998.

12.2 ROBUST CONTROL SYSTEMS AND SYSTEM SENSITIVITY

Designing highly accurate systems in the presence of significant plant uncertainty is a classical feedback design problem. The theoretical bases for the solution of this problem date back to the works of H. S. Black and H. W. Bode in the early 1930s, when this problem was referred to as the sensitivities design problem. A significant amount of literature has been published since then regarding the design of systems subject to large plant uncertainty. The designer seeks to obtain a system that performs adequately over a large range of uncertain parameters. A system is said to be robust when it is durable, hardy, and resilient.

A control system is robust when (1) it has low sensitivities, (2) it is stable over the range of parameter variations, and (3) the performance continues to meet the specifications in the presence of a set of changes in the system parameters [3, 4]. Robustness is the sensitivity to effects that are not considered in the analysis and design phase—for example, disturbances, measurement noise, and unmodeled dynamics. The system should be able to withstand these neglected effects when performing the tasks for which it was designed.

For small-parameter perturbations we may use, as a measure of robustness, the differential sensitivities discussed in Sections 4.2 (system sensitivity) and 7.6 (root sensitivity) [6].

System sensitivity is defined as

$$S_{\alpha}^T = \frac{\partial T/T}{\partial \alpha/\alpha}, \quad (12.1)$$

where α is the parameter, and T the transfer function of the system.

Root sensitivity is defined as

$$S_{\alpha}^{r_i} = \frac{\partial r_i}{\partial \alpha/\alpha}. \quad (12.2)$$

When the zeros of $T(s)$ are independent of the parameter α , we showed that

$$S_{\alpha}^T = -\sum_{i=1}^n S_{\alpha}^{r_i} \cdot \frac{1}{(s + r_i)}, \quad (12.3)$$

for an n th-order system. For example, if we have a closed-loop system, as shown in Fig. 12.2, where the variable parameter is α , then $T(s) = 1/[s + (\alpha + 1)]$, and

$$S_{\alpha}^T = \frac{-\alpha}{s + \alpha + 1}. \quad (12.4)$$

Furthermore the root is $r_1 = +(\alpha + 1)$, and

$$-S_{\alpha}^{r_1} = -\alpha. \quad (12.5)$$

FIGURE 12.2
A first-order system.

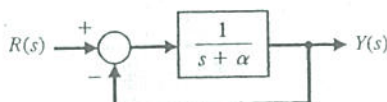


FIGURE 12.1
A second-order system.

FIGURE 12.3
Sensitivity of a first-order system is shown.

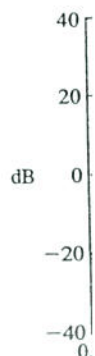
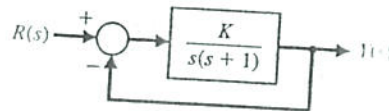


FIGURE 12.3
A second-order system.



Therefore

$$S_a^T = -S_a^{r_1} \frac{1}{(s + \alpha + 1)}. \quad (12.6)$$

Let us examine the sensitivity of the second-order system shown in Fig. 12.3. The transfer function of the closed-loop system is

$$T(s) = \frac{K}{s^2 + s + K}. \quad (12.7)$$

As we saw in Eq. (4.12), the system sensitivity for K is

$$S_K^T = \frac{1}{1 + GH(s)} = \frac{s(s+1)}{s^2 + s + K}. \quad (12.8)$$

A Bode plot of the asymptotes of $20 \log |T(j\omega)|$ and $20 \log |S(j\omega)|$ is shown in Fig. 12.4 for $K = 1/4$ (critical damping). Note that the sensitivity is small for lower frequencies, while the transfer function primarily passes low frequencies. Also note for this case, $T(s) = 1 - S(s)$.

Of course the sensitivity S only represents robustness for small changes in gain. If K changes from $1/4$ within the range $K = 1/16$ to $K = 1$, the resulting range of step response is shown in Fig. 12.5. This system, with an expected wide range of K , may not be considered adequately robust. A robust system would be expected to yield essentially the same (within an agreed-upon variation) response to a selected input.

FIGURE 12.4
Sensitivity and $20 \log |T(j\omega)|$ for the second-order system in Fig. 12.3. The asymptotic approximations are shown for $K = 1/4$.

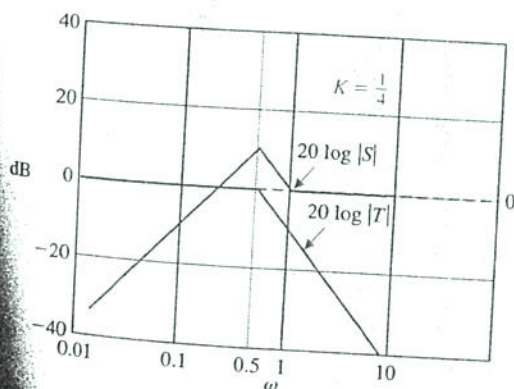


FIGURE 12.5
Step response for selected gain K .

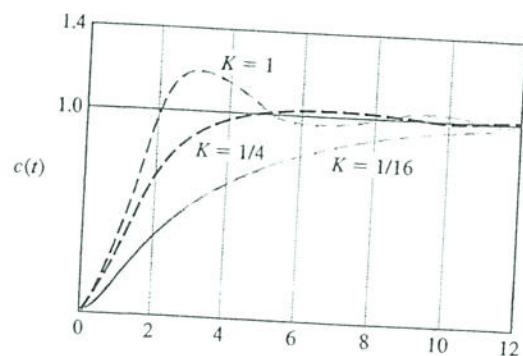
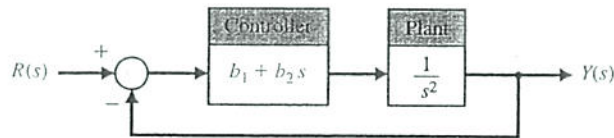


FIGURE 12.6
A system with a
PD controller.



EXAMPLE 12.1 Sensitivity of a controlled system

Consider the system shown in Fig. 12.6, where $G(s) = 1/s^2$, and a PD controller $G_c(s) = b_1 + b_2s$. Then the sensitivity with respect to changes in $G(s)$ is

$$S_G^T = \frac{1}{1 + GG_c(s)} = \frac{s^2}{s^2 + b_2s + b_1}, \quad (12.9)$$

and

$$T(s) = \frac{b_2s + b_1}{s^2 + b_2s + b_1}. \quad (12.10)$$

Consider the normal condition $\zeta = 1$ and $\omega_n = \sqrt{b_1}$. Then $b_2 = 2\omega_n$ to achieve $\zeta = 1$. Therefore we may plot $20 \log |S|$ and $20 \log |T|$ on a Bode diagram, as shown in Fig. 12.7. Note that the frequency ω_n is an indicator on the boundary between the frequency region in which the sensitivity is the important design criterion and the region in which the stability margin is important. Thus if we specify ω_n properly to take into consideration the extent of modeling error and the frequency of external disturbance, we can expect the system to have an acceptable amount of robustness. Note that $G_c(s)$ is a proportional-derivative (PD) controller. ■

EXAMPLE 12.2 System with a right-hand-plane zero

Consider the system shown in Fig. 12.8, where the plant has a zero in the right-hand plane. The closed-loop transfer function is

$$T(s) = \frac{K(s - 1)}{s^2 + (2 + K)s + (1 - K)}. \quad (12.11)$$

FIGURE 12.7
Sensitivity and
 $T(s)$ for the
second-order
system in
Fig. 12.6.

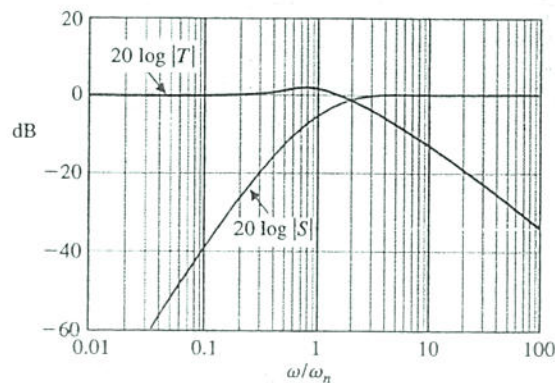


FIGURE 12.8
A second-order system.

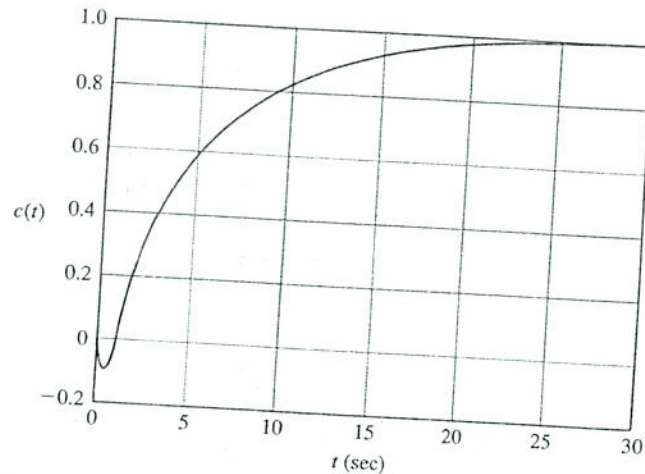
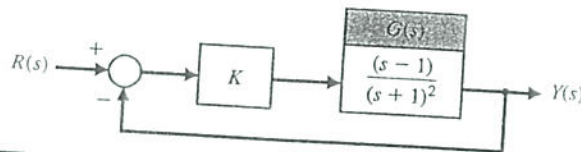


FIGURE 12.9
Step response of the system in Fig. 12.8 with $K = 1/2$.

The system is stable for a gain $-2 < K < 1$. The steady-state error is

$$e_{ss} = \frac{1 - 2K}{1 - K}, \quad (12.12)$$

and $e_{ss} = 0$ when $K = 1/2$. The response to a negative unit step input $R(s) = -1/s$ is shown in Fig. 12.9. Note the initial undershoot at $t = 1$ second. This system is sensitive to changes in K , as recorded in Table 12.1. The performance of this system might be considered barely acceptable for a change of gain of only $\pm 10\%$. Thus this system would not be considered robust. The steady-state error of this system changes greatly as K changes. ■

12.3 ANALYSIS OF ROBUSTNESS

Consider the closed-loop system shown in Fig. 12.1. System goals include maintaining small tracking error $[e(t) = r(t) - y(t)]$ for an input $r(t)$ and keeping the output $y(t)$ small

TABLE 12.1 Results for Example 12.2

K	0.25	0.45	0.50	0.55	0.75
$ e_{ss} $	0.67	0.18	0	0.22	1.0
Undershoot	5%	9%	10%	11%	15%
Settling time (seconds)	15	24	27	30	45

for a disturbance $d(t)$. The sensor noise $n(t)$ must be small with respect to $r(t)$ so that $|r| \gg |n|$.

The sensitivity function is

$$S(s) = [1 + G_c(s)G(s)]^{-1},$$

and the closed-loop transfer function is

$$T(s) = \frac{G_c(s)G(s)}{1 + G_c(s)G(s)}$$

when $G_p(s) = 1$. We then have

$$S(s) + T(s) = 1. \quad (12.13)$$

Thus it is important to make $S(s)$ small. For physically realizable systems, the loop gain $L(s) = G_c(s)G(s)$ must be small for high frequencies. This means that $S(j\omega)$ approaches 1 at high frequencies.

An **additive perturbation** characterizes the set of possible plants as follows (here we assume $G_c(s) = 1$):

$$G_a(s) = G(s) + A(s),$$

where $G(s)$ is the nominal plant, and $A(s)$ is the perturbation that is bounded in magnitude. It is assumed that $G_a(s)$ and $G(s)$ have the same number of poles in the right-hand s -plane (if any) [36]. Then the system stability will not change if

$$|A(j\omega)| < |1 + G(j\omega)| \quad \text{for all } \omega. \quad (12.14)$$

This assures stability but not dynamic performance.

A **multiplicative perturbation** gives the plant

$$G_m(s) = G(s)[1 + M(s)].$$

The perturbation is bounded in magnitude, and again it is assumed that $G_m(s)$ and $G(s)$ have the same number of poles in the right-hand s -plane. Then the system stability will not change if

$$|M(j\omega)| < \left| 1 + \frac{1}{G(j\omega)} \right| \quad \text{for all } \omega. \quad (12.15)$$

Equation (12.15) is called the **robust stability criterion**. This is a test for robustness with respect to a multiplicative perturbation. This form of perturbation is often used because it satisfies the intuitive properties of (1) being small at low frequencies, where the nominal plant model is usually well known, and (2) being large at high frequencies, where the nominal model is always inexact.

EXAMPLE 12.3 System with multiplicative perturbation

Consider the system of Fig. 12.1 with $G_p(s) = 1$, $G_c = K$, and

$$G(s) = \frac{170,000(s + 0.1)}{s(s + 3)(s^2 + 10s + 10,000)}.$$

FIGURE 12
The robust s
criterion for
Example 12

The system is unstable with $K = 1$, but a reduction in gain to $K = 0.5$ will stabilize it. Now consider the effect of an unmodeled pole at 50 rad/s. In this case, the multiplicative perturbation is determined from

$$[1 + M(s)] = \frac{50}{s + 50},$$

or $M(s) = -s/(s + 50)$. The magnitude bound is then

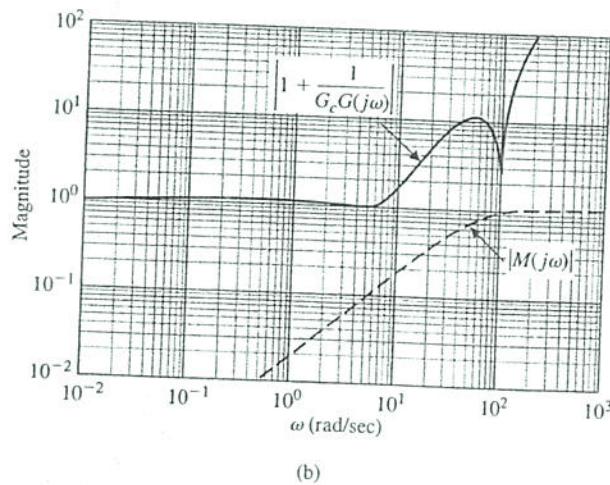
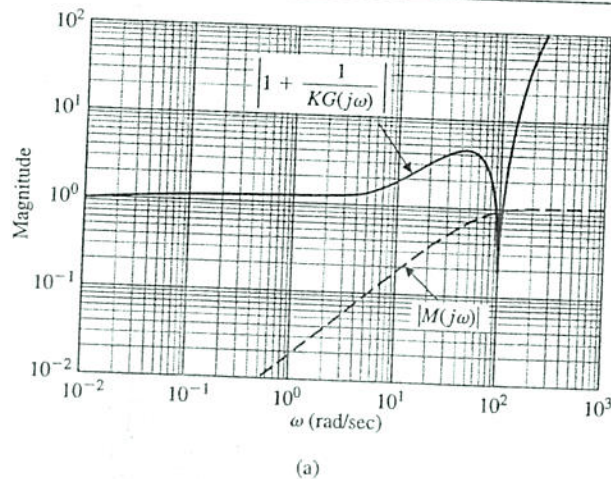
$$|M(j\omega)| = \left| \frac{-j\omega}{j\omega + 50} \right|.$$

$|M(j\omega)|$ and $|1 + 1/KG(j\omega)|$ are plotted in Fig. 12.10(a), where it is seen that the criterion of Eq. (12.15) is not satisfied. Thus the system may not be stable.

If we use a lag compensator $G_c(s)$, where

$$G_c(s) = \frac{0.15(s + 25)}{(s + 2.5)},$$

FIGURE 12.10
The robust stability
criterion for
Example 12.3.



the loop transfer function is $1 + G_c(s)G(s)$, and we reshape the function $G_cG(j\omega)$ in the frequency range $2 < \omega < 25$. Then we have the altered magnitude

$$\left| 1 + \frac{1}{G_cG(j\omega)} \right|,$$

as plotted in Fig. 12.10(b). Here the robustness inequality is satisfied, and the system remains stable. ■

The control objective is to select a prefilter $G_p(s)$ and a compensator $G_c(s)$ in a two-degree-of-freedom feedback configuration (Fig. 12.1) so that the transient, steady-state, and frequency-domain specifications are achieved and the cost of feedback measured by the bandwidth of the compensator $G_c(j\omega)$ is sufficiently small. This bandwidth constraint is needed mainly because of noise that is inevitable in measuring the system output. A large noise amplification can saturate either the latter stages of $G_c(s)$ or the early plant stages.

EXAMPI

12.4 SYSTEMS WITH UNCERTAIN PARAMETERS

Many systems have several parameters that are constants but uncertain within a range. For example, consider a system with a characteristic equation

$$s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \cdots + a_0 = 0 \quad (12.16)$$

with known coefficients within bounds

$$\alpha_i \leq a_i \leq \beta_i, \quad i = 0, \dots, n,$$

where $a_n = 1$.

To ascertain the stability of the system, one might have to investigate all possible combinations of parameters. Fortunately it is possible to investigate a limited number of worst-case polynomials [22]. The analysis of only four polynomials is sufficient, and they are readily defined for a third-order system with a characteristic equation

$$s^3 + a_2s^2 + a_1s + a_0 = 0. \quad (12.17)$$

Then the four polynomials are

$$q_1(s) = s^3 + \alpha_2s^2 + \beta_1s + \beta_0,$$

$$q_2(s) = s^3 + \beta_2s^2 + \alpha_1s + \alpha_0,$$

$$q_3(s) = s^3 + \beta_2s^2 + \beta_1s + \alpha_0,$$

$$q_4(s) = s^3 + \alpha_2s^2 + \alpha_1s + \beta_0.$$

One of the four polynomials represents the **worst case** and may indicate unstable performance or at least the worst performance for the system in that case.

EXAMPLE 12.4 Third-order system with uncertain coefficients

Consider a third-order system with uncertain coefficients such that

$$8 \leq a_0 \leq 60 \Rightarrow \alpha_0 = 8, \quad \beta_0 = 60;$$

$$12 \leq a_1 \leq 100 \Rightarrow \alpha_1 = 12, \quad \beta_1 = 100;$$

$$7 \leq a_2 \leq 25 \Rightarrow \alpha_2 = 7, \quad \beta_2 = 25.$$

The four polynomials are

$$q_1(s) = s^3 + 7s^2 + 100s + 60,$$

$$q_2(s) = s^3 + 25s^2 + 12s + 8,$$

$$q_3(s) = s^3 + 25s^2 + 100s + 8,$$

$$q_4(s) = s^3 + 7s^2 + 12s + 60.$$

We then proceed to check these four polynomials using the Routh-Hurwitz criterion and hence determine that the system is stable for all the range of uncertain parameters. ■

EXAMPLE 12.5 Stability of uncertain system

Consider a unity feedback system with a plant transfer function (nominal conditions)

$$G(s) = \frac{4.5}{s(s+1)(s+2)}.$$

The nominal characteristic equation is then

$$q(s) = s^3 + 3s^2 + 2s + 4.5 = 0.$$

Using the Routh-Hurwitz criterion, we find that this system is nominally stable. However, if the system has uncertain coefficients such that

$$4 \leq a_0 \leq 5 \Rightarrow \alpha_0 = 4, \quad \beta_0 = 5;$$

$$1 \leq a_1 \leq 3 \Rightarrow \alpha_1 = 1, \quad \beta_1 = 3; \text{ and}$$

$$2 \leq a_2 \leq 4 \Rightarrow \alpha_2 = 2, \quad \beta_2 = 4,$$

then we must examine the four polynomials:

$$q_1(s) = s^3 + 2s^2 + 3s + 5,$$

$$q_2(s) = s^3 + 4s^2 + 1s + 4,$$

$$q_3(s) = s^3 + 4s^2 + 3s + 4,$$

$$q_4(s) = s^3 + 2s^2 + 1s + 5.$$

Using the Routh-Hurwitz criterion, $q_1(s)$ and $q_3(s)$ are stable and $q_2(s)$ is marginally stable. For $q_4(s)$ we have

$$\begin{array}{c|cc} s^3 & 1 & 1 \\ s^2 & 2 & 5 \\ s^1 & -3/2 & \\ s^0 & 5 & \end{array}$$

Therefore the system is unstable for the worst case, where $\alpha_2 = \text{minimum}$, $\alpha_1 = \text{minimum}$, and $\beta_0 = \text{maximum}$. This occurs when the plant has changed to

$$G(s) = \frac{5}{s(s+1)(s+1)}.$$

Note that the third pole has moved toward the $j\omega$ -axis to its limit at $s = -1$ and that the gain has increased to its limit at $K = 5$. Often we are able to examine the transfer function $G(s)$ and predict the worst-case conditions. ■

12.5 THE DESIGN OF ROBUST CONTROL SYSTEMS

The design of robust control systems is based on two tasks: determining the structure of the controller and adjusting the controller's parameters to give an "optimal" system performance. This design process is normally done with "assumed complete knowledge" of the plant. Furthermore the plant is normally described by a linear time-invariant continuous model. The structure of the controller is chosen such that the system's response can meet certain performance criteria.

One possible objective in the design of a control system is that the controlled system's output should exactly and instantaneously reproduce its input. That is, the system's transfer function should be unity:

$$T(s) = \frac{Y(s)}{R(s)} = 1. \quad (12.18)$$

In other words, the system should be presentable on a Bode gain versus frequency diagram with a 0-dB gain of infinite bandwidth and zero phase shift. In practice this is not possible, since every system will contain inductive- and capacitive-type components that store energy in some form. It is these elements and their interconnections with energy-dissipative components that produce the system's dynamic response characteristics. Such systems reproduce some inputs almost exactly, while other inputs are not reproduced at all, signifying that the system's bandwidth is less than infinite.

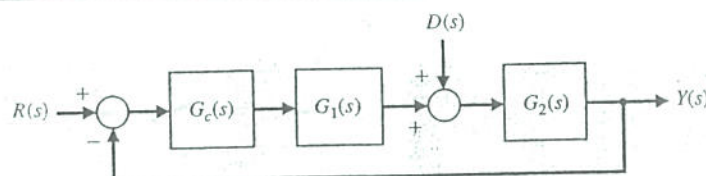
Once we recognize that the system's dynamics cannot be ignored, we need a new design objective. One possible design objective is to maintain the magnitude response curve as flat and as close to unity for as large a bandwidth as possible for a given plant and controller combination [22].

Another important goal of a control system design is that the effect on the output of the system due to disturbances is minimized. Thus we wish to minimize $Y(s)/D(s)$ over a range of frequency.

Consider the control system shown in Fig. 12.11, where $G(s) = G_1(s)G_2(s)$ is the plant, and $D(s)$ is the disturbance. We then have

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G_c G_1 G_2(s)}{1 + G_c G_1 G_2(s)}, \quad (12.19)$$

FIGURE 12.11
A system with a disturbance.



and

$$\frac{Y(s)}{D(s)} = \frac{G_2(s)}{1 + G_c G_1 G_2(s)} \quad (12.20)$$

Note that both the reference and disturbance transfer functions have the same denominator; in other words, they have the same characteristic equation—that is,

$$1 + G_c(s)G_1(s)G_2(s) = 1 + L(s) = 0. \quad (12.21)$$

Furthermore we recall that the sensitivity of $T(s)$ with respect to $G(s)$ is

$$S_G^T = \frac{1}{1 + G_c G_1 G_2(s)}, \quad (12.22)$$

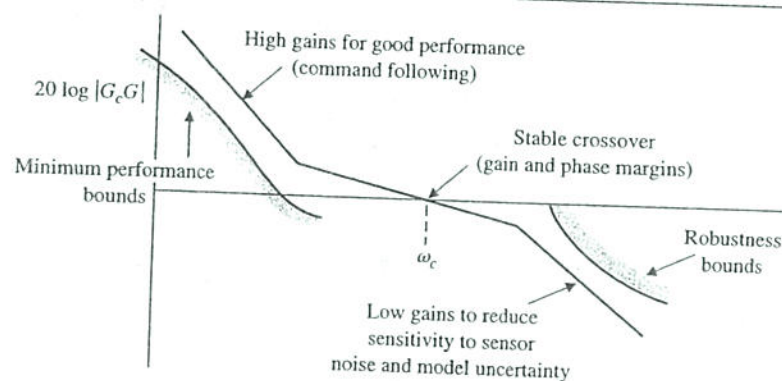
and the characteristic equation is the influencing factor on the sensitivity. Equation (12.22) shows that for low sensitivity S we require a high value of loop gain $L(j\omega)$, but it is known that high gain could cause instability or poor responsiveness of $T(s)$. Thus the designer seeks the following:

1. $T(s)$ with wide bandwidth and faithful reproduction of $R(s)$.
2. Large loop gain $L(s)$ in order to minimize sensitivity S .
3. Large loop gain $L(s)$ attained primarily by $G_c(s)G_1(s)$, since $Y(s)/D(s) \approx 1/G_c G_1(s)$.

Setting the design of robust systems in frequency-domain terms, we are required to find a proper compensator, $G_c(s)$, such that the closed-loop sensitivity is less than some tolerance value, and sensitivity minimization involves finding a proper compensator such that the closed-loop sensitivity equals or is arbitrarily close to the minimal attainable sensitivity. Similarly, the gain margin problem is to find a proper compensator to achieve some prescribed gain margin, and gain margin maximization involves finding a proper compensator to achieve the maximal attainable gain margin. For the frequency-domain specifications, we require the following for the Bode diagram of $G_c G(j\omega)$, shown in Fig. 12.12:

1. For relative stability, $G_c G(j\omega)$ must have, for an adequate range of ω , not more than -20 dB/decade slope at or near the crossover frequency ω_c .
2. Steady-state accuracy achieved by the low frequency gain.

FIGURE 12.12
Bode diagram for
 $20 \log |G_c G(j\omega)|$.



3. Accuracy over a bandwidth ω_B by not allowing $|G_c G(\omega)|$ to fall below a prescribed level.
4. Disturbance rejection by high gain for $G_c(j\omega)$ over the system bandwidth.

Using the root sensitivity concept, we can state that we require that S_a^r be minimized while attaining $T(s)$ with dominant roots that will provide the appropriate response and minimize the effect of $D(s)$. Again we see that the goal is to have the gain of the loop primarily attained by $G_c(s)$. As an example, let $G_c(s) = K$, $G_1(s) = 1$, and $G_2(s) = 1/s(s+1)$ for the system in Fig. 12.11. This system has two roots, and we select a gain K so that $Y(s)/D(s)$ is minimized, S_K^r is minimized, and $T(s)$ has desirable dominant roots. The sensitivity is

$$S_K^r = \frac{dr}{dK} \cdot \frac{K}{r} = \frac{ds}{dK} \bigg|_{s=r} \cdot \frac{K}{r}, \quad (12.23)$$

and the characteristic equation is

$$s(s+1) + K = 0. \quad (12.24)$$

Therefore $dK/ds = -(2s+1)$, since $K = -s(s+1)$. We then obtain

$$S_K^r = \frac{-1}{(2s+1)} \cdot \frac{[-s(s+1)]}{s} \bigg|_{s=r}. \quad (12.25)$$

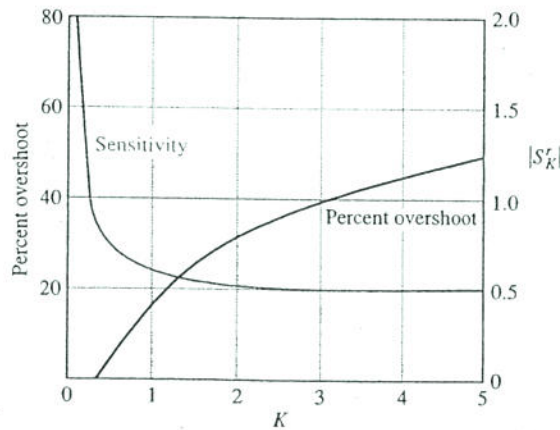
When $\zeta < 1$, the roots are complex, and $r = -0.5 + j\omega$. Then,

$$|S_K^r| = \left(\frac{0.25 + \omega^2}{4\omega^2} \right)^{1/2}. \quad (12.26)$$

The magnitude of the sensitivity is plotted in Fig. 12.13 for $K = 0.2$ to $K = 5$. The percent overshoot to a step is also shown. It is best to reduce the sensitivity but to limit K to 1.5 or less. We then attain the majority of the attainable reduction in sensitivity while maintaining good performance for the step response. In general we can use the design procedure as follows:

1. Sketch the root locus of the compensated system with $G_c(s)$ chosen to attain the desired location for the dominant roots.

FIGURE 12.13
Sensitivity and
percent overshoot
for a second-order
system.



2. Maximize the gain of $G_c(s)$ to reduce the effect of the disturbance.
3. Determine S_α^r and attain the minimum value of the sensitivity consistent with the transient response required, as described in step 1.

EXAMPLE 12.6 Sensitivity and compensation

Let us reconsider the system in Example 10.1 when $G(s) = 1/s^2$, $H(s) = 1$, and $G_c(s)$ is to be selected by frequency response methods. Therefore the compensator is to be selected to achieve an appropriate gain and phase margin while minimizing sensitivity and the effect of the disturbance. Thus we choose

$$G_c(s) = \frac{K(s/z + 1)}{(s/p + 1)} \quad (12.27)$$

As in Example 10.1, we choose $K = 10$ to reduce the effect of the disturbance. To attain a phase margin of 45° , we select $z = 2.0$ and $p = 12.0$. We then attain the compensated diagram shown in Fig. 10.9 and repeated in Fig. 12.14. Recall that the closed-loop bandwidth is $\omega_B = 1.6 \omega_c$. Thus we will increase the bandwidth by using the compensator and improve the fidelity of reproduction of the input signals.

The sensitivity may be ascertained at ω_c as

$$|S_G^T(\omega_c)| = \left| \frac{1}{1 + G_c G(j\omega)} \right|_{\omega_c} \quad (12.28)$$

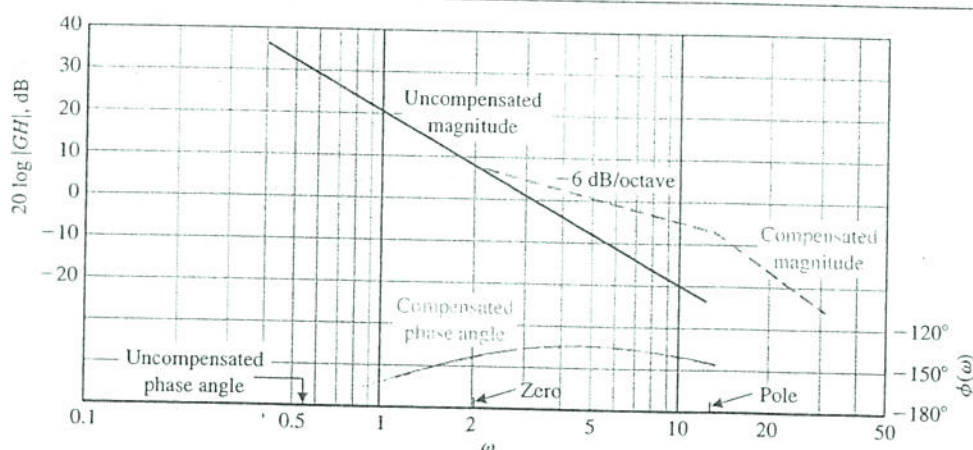
To estimate $|S_G^T|$, we recall that the Nichols chart enables us to obtain

$$|T(\omega)| = \left| \frac{G_c G(\omega)}{1 + G_c G(\omega)} \right| \quad (12.29)$$

Thus we can plot a few points of $G_c G(j\omega)$ on the Nichols chart and then read $T(\omega)$ from the chart. Then

$$|S_G^T(\omega_1)| = \frac{|T(\omega_1)|}{|G_c G(\omega_1)|} \quad (12.30)$$

FIGURE 12.14
Bode diagram for
Example 12.6.



where ω_1 is chosen arbitrarily as $\omega_c/2.5$. In general we choose a frequency below ω_c to determine the value of $|S(\omega_1)|$. Of course we desire a low value of sensitivity. The Nichols chart for the compensated system is shown in Fig. 12.15. For $\omega_1 = \omega_c/2.5 = 2$, we have $20 \log T = 2.5 \text{ dB}$ and $20 \log G_c G = 9 \text{ dB}$. Therefore

$$|S(\omega_1)| = \frac{|T|}{|G_c G|} = \frac{1.33}{2.8} = 0.47. \blacksquare$$

FIGURE 12.15
Nichols chart for
Example 12.7.

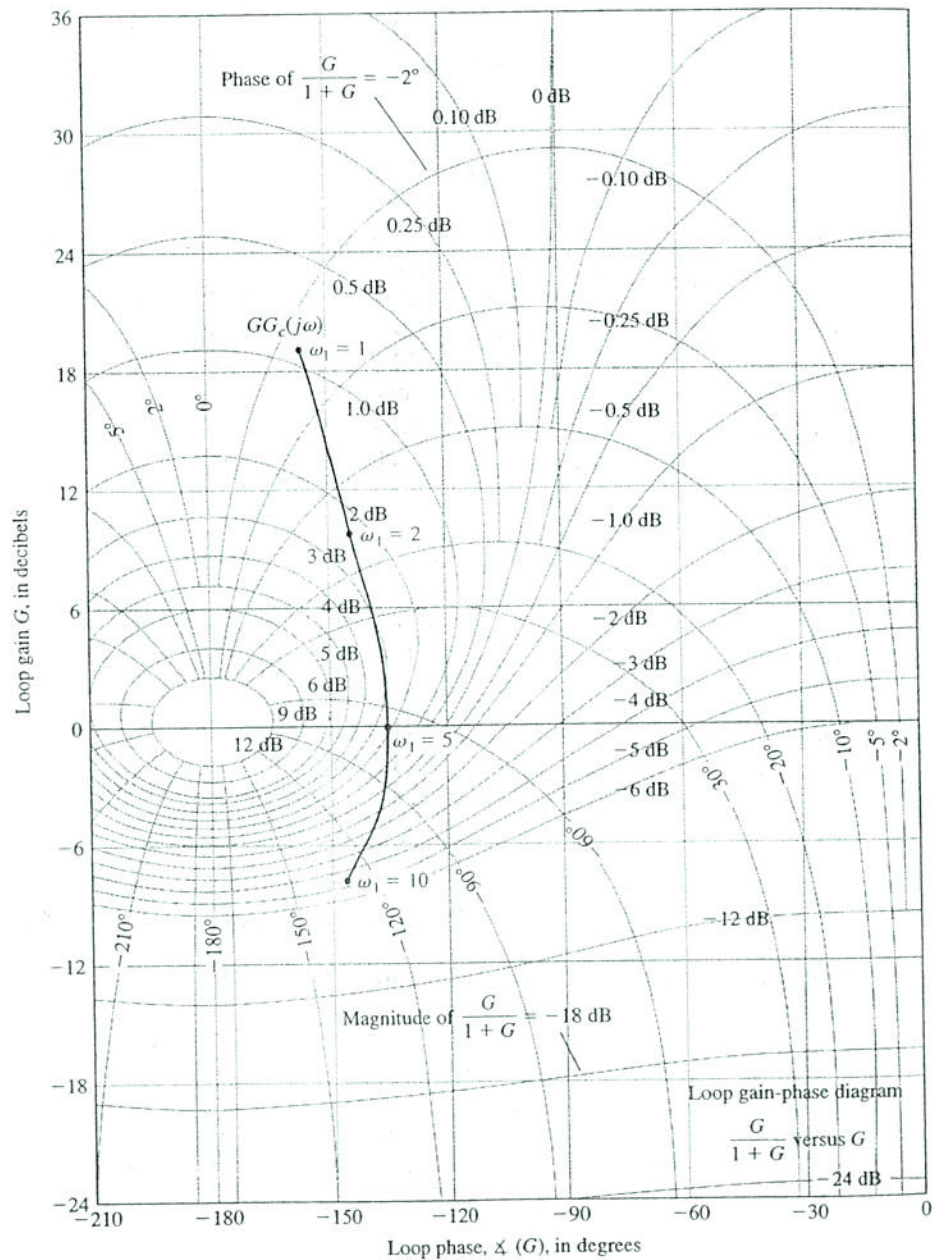
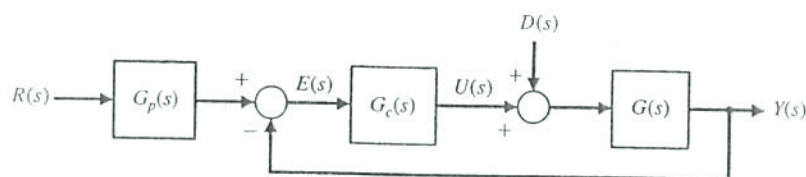


FIGURE 12.16
Feedback control
system with a
desired input $R(s)$
and an undesired
input $D(s)$.



EXAMPLE 12.7 Sensitivity with a lead compensator

Let us again consider the system in Example 12.6, using the root locus design obtained in Example 10.3. The compensator was chosen as

$$G_c(s) = \frac{8.1(s + 1)}{(s + 3.6)}, \quad (12.31)$$

for the system of Fig. 12.16. The dominant roots are thus $s = -1 \pm j2$. Since the gain is 8.1, the effect of the disturbance is reduced, and the time response meets the specifications. The sensitivity at a root r may be obtained by assuming that the system, with dominant roots, may be approximated by the second-order system

$$T(s) = \frac{K}{s^2 + 2\zeta\omega_n s + K} = \frac{K}{s^2 + 2s + K},$$

since $\zeta\omega_n = 1$. The characteristic equation is thus

$$s^2 + 2s + K = 0.$$

Then $dK/ds = -(2s + 2)$, since $K = -(s^2 + 2s)$. Therefore

$$S_K^r = \frac{-1}{(2s + 2)} \cdot \frac{-(s^2 + 2s)}{s} \bigg|_{s=r} = \frac{(s + 2)}{(2s + 2)} \bigg|_{s=r}, \quad (12.32)$$

where $r = -1 + j2$. Then substituting $s = r$, we obtain

$$|S_K^r| = 0.56.$$

If we raise the gain to $K = 10$, we expect $r \cong -1.1 \pm j2.2$. Then the sensitivity is

$$|S_K^r| = 0.53.$$

Thus as K increases, the sensitivity decreases, but the transient performance deteriorates. ■

12.6 PID CONTROLLERS

The PID controller has the transfer function

$$G_c(s) = K_1 + \frac{K_2}{s} + K_3 s.$$

The popularity of PID controllers can be attributed partly to their robust performance in a wide range of operating conditions and partly to their functional simplicity, which allows

engineers to operate them in a simple, straightforward manner. To implement such a controller, three parameters must be determined for the given process: proportional gain, integral gain, and derivative gain [35].

Consider the PID controller

$$\begin{aligned} G_c(s) &= K_1 + \frac{K_2}{s} + K_3 s = \frac{K_3 s^2 + K_1 s + K_2}{s} \\ &= \frac{K_3(s^2 + as + b)}{s} = \frac{K_3(s + z_1)(s + z_2)}{s}, \end{aligned} \quad (12.33)$$

where $a = K_1/K_3$, and $b = K_2/K_3$. Therefore a PID controller introduces a transfer function with one pole at the origin and two zeros that can be located anywhere in the left-hand s -plane.

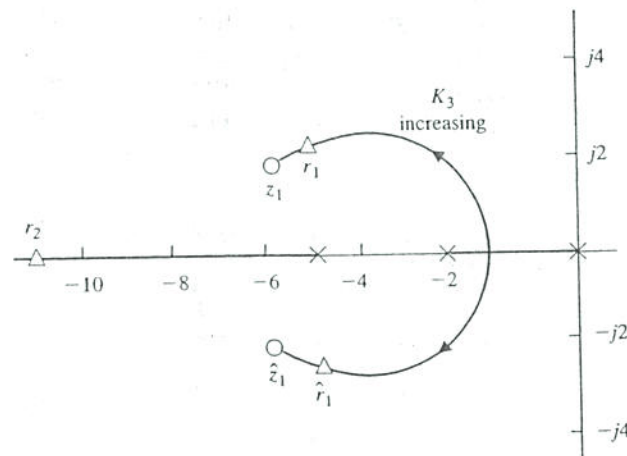
Recall that a root locus begins at the poles and ends at the zeros. If we have a system as shown in Fig. 12.16 with

$$G(s) = \frac{1}{(s + 2)(s + 5)},$$

and we use a PID controller with complex zeros, we can plot the root locus as shown in Fig. 12.17. As the gain, K_3 , of the controller is increased, the complex roots approach the zeros. The closed-loop transfer function is

$$\begin{aligned} T(s) &= \frac{G(s)G_c(s)G_p(s)}{1 + G(s)G_c(s)} \\ &= \frac{K_3(s + z_1)(s + \hat{z}_1)}{(s + r_2)(s + r_1)(s + \hat{r}_1)} G_p(s) \\ &\cong \frac{K_3 G_p(s)}{(s + r_2)}, \end{aligned} \quad (12.34)$$

FIGURE 12.17
Root locus with
 $z_1 = -6 + j2$.



because the zeros and the complex roots are approximately equal ($r_1 \approx z_1$). Setting $G_p(s) = 1$, we have

$$T(s) = \frac{K_3}{s + r_2} \approx \frac{K_3}{s + K_3} \quad (12.35)$$

when $K_3 \gg 1$. The only limiting factor is the allowable magnitude of $U(s)$ (Fig. 12.16) when K_3 is large. If K_3 is 100, the system has a fast response and zero steady-state error. Furthermore, the effect of the disturbance is reduced significantly.

In general we note that PID controllers are particularly useful for reducing steady-state error and improving the transient response when $G(s)$ has one or two poles (or may be approximated by a second-order plant).

12.7 THE DESIGN OF ROBUST PID-CONTROLLED SYSTEMS

The selection of the three coefficients of PID controllers is basically a search problem in a three-dimensional space. Points in the search space correspond to different selections of a PID controller's three parameters. By choosing different points of the parameter space, we can produce, for example, different step responses for a step input. A PID controller can be determined by moving in this search space on a trial-and-error basis.

The main problem in the selection of the three coefficients is that these coefficients do not readily translate into the desired performance and robustness characteristics that the control system designer has in mind. Several rules and methods have been proposed to solve this problem. In this section we consider several design methods using root locus and performance indices.

The first design method uses the ITAE performance index of Section 5.9 and the optimum coefficients of Table 5.6 for a step input or Table 5.7 for a ramp input. Hence we select the three PID coefficients to minimize the ITAE performance index, which produces an excellent transient response to a step (Fig. 5.30c) or a ramp. The design procedure consists of three steps:

1. Select the ω_n of the closed-loop system by specifying the settling time.
2. Determine the three coefficients using the appropriate optimum equation (Table 5.6) and the ω_n of step 1 to obtain $G_c(s)$.
3. Determine a prefilter $G_p(s)$ so that the closed-loop system transfer function, $T(s)$, does not have any zeros, as required by Eq. (5.47).

EXAMPLE 12.8 Robust control of temperature

Consider a temperature controller with a control system as shown in Fig. 12.16 and a plant

$$G(s) = \frac{1}{(s + 1)^2}. \quad (12.36)$$

If $G_c(s) = 1$, the steady-state error is 50%, and the settling time (2% criterion) is 3.2 seconds for a step input. We desire to obtain an optimum ITAE performance for a step input and a settling time of less than 0.5 second. Using a PID controller we have

$$G_c(s) = \frac{K_3 s^2 + K_1 s + K_2}{s}. \quad (12.37)$$

Therefore the closed-loop transfer function without prefiltering [$G_p(s) = 1$] is

$$T_1(s) = \frac{Y(s)}{R(s)} = \frac{G_c G(s)}{1 + G_c G(s)} \quad (12.38)$$

$$= \frac{K_3 s^2 + K_1 s + K_2}{s^3 + (2 + K_3)s^2 + (1 + K_1)s + K_2}$$

The optimum coefficients of the characteristic equation for ITAE are obtained from Table 5.6 as

$$(s^3 + 1.75\omega_n s^2 + 2.15\omega_n^2 s + \omega_n^3). \quad (12.39)$$

We need to select ω_n in order to meet the settling time requirement. Since $T_s = 4/\zeta\omega_n$ and ζ is unknown but near 0.8, we set $\omega_n = 10$. Then equating the denominator of Eq. (12.38) to Eq. (12.39), we obtain the three coefficients as $K_1 = 214$, $K_3 = 15.5$, and $K_2 = 1000$.

Then Eq. (12.38) becomes

$$T_1(s) = \frac{15.5s^2 + 214s + 1000}{s^3 + 17.5s^2 + 215s + 1000} \quad (12.40)$$

$$= \frac{15.5(s + 6.9 + j4.1)(s + 6.9 - j4.1)}{s^3 + 17.5s^2 + 215s + 1000}$$

The response of this system to a step input has an overshoot of 32%, as recorded in Table 12.2.

We select a prefilter $G_p(s)$ so that we achieve the desired ITAE response with

$$T(s) = \frac{G_c G G_p(s)}{1 + G_c G(s)} = \frac{1000}{s^3 + 17.5s^2 + 215s + 1000} \quad (12.41)$$

Therefore we require

$$G_p(s) = \frac{64.5}{(s^2 + 13.8s + 64.5)} \quad (12.42)$$

in order to eliminate the zeros in Eq. (12.40) and bring the overall numerator to 1000. The response of the system, $T(s)$, to a step input is indicated in Table 12.2. The system has a small overshoot, a settling time of less than $\frac{1}{2}$ second, and zero steady-state error. Furthermore for a disturbance $D(s) = 1/s$, the maximum value of $y(t)$ due to the disturbance is 0.4% of the magnitude of the disturbance. This is a very favorable design. ■

TABLE 12.2 Results for Example 12.8

Controller	$G_c(s) = 1$	PID and $G_p(s) = 1$	PID with $G_p(s)$ Prefilter
Percent overshoot	0	31.7%	1.9%
Settling time (seconds)	3.2	0.20	0.45
Steady-state error	50.1%	0.0%	0.00%
$ y(t)/d(t) _{\text{maximum}}$	52%	0.4%	0.4%

EXAMPLE 12.9 Robust system design

Let us reconsider the system in Example 12.8 when the plant varies significantly, so that

$$G(s) = \frac{K}{(\tau s + 1)^2}, \quad (12.43)$$

where $0.5 \leq \tau \leq 1$, and $1 \leq K \leq 2$. It is desired to achieve robust behavior using an ITAE optimum system with a prefilter while attaining an overshoot of less than 4% and a settling time (2% criterion) of less than 2 seconds, while $G(s)$ can attain any value in the range indicated. We select $\omega_n = 8$ in order to attain the settling time and determine the ITAE coefficients for $K = 1$ and $\tau = 1$. Completing the calculation, we obtain the system without a prefilter [$G_p(s) = 1$] as

$$T_1(s) = \frac{12(s^2 + 11.38s + 42.67)}{s^3 + 14s^2 + 137.6s + 512}, \quad (12.44)$$

and

$$G_c(s) = \frac{12(s^2 + 11.38s + 42.67)}{s}. \quad (12.45)$$

We select a prefilter

$$G_p(s) = \frac{42.67}{(s^2 + 11.38s + 42.67)} \quad (12.46)$$

to obtain the optimum ITAE transfer function

$$T(s) = \frac{512}{s^3 + 14s^2 + 137.6s + 512}. \quad (12.47)$$

We then obtain the step response for the four conditions: $\tau = 1, K = 1$; $\tau = 0.5, K = 1$; $\tau = 1, K = 2$; and $\tau = 0.5, K = 2$. The results are summarized in Table 12.3. This is a very robust system. ■

The value of ω_n that can be chosen will be limited by considering the maximum allowable $u(t)$, where $u(t)$ is the output of the controller, as shown in Fig. 12.16. If the maximum value of $e(t)$ is 1, then $u(t)$ would normally be limited to 100 or less. As an example, consider the system in Fig. 12.16 with a PID controller, $G(s) = 1/s(s + 1)$, and the necessary prefilter $G_p(s)$ to achieve ITAE performance. If we select $\omega_n = 10, 20$, and 40 , the maximum value of $u(t)$ is as recorded in Table 12.4. If we wish to limit $u(t)$ to a maximum equal to 100, we need to limit ω_n to 16. Thus we are limited in the settling time we can achieve.

TABLE 12.3 Results for Example 12.9 with $\omega_n = 8$

Plant Conditions	$\tau = 1$ $K = 1$	$\tau = 0.5,$ $K = 1$	$\tau = 1,$ $K = 2$	$\tau = 0.5,$ $K = 2$
Percent overshoot	2%	0%	0%	1%
Settling time (seconds)	1.25	0.8	0.8	0.9

TABLE 12.4 Maximum Value of Plant Input

ω_n	10	20	40
$u(t)$ maximum for $R(s) = 1/s$	35	135	550
Settling time (seconds)	0.9	0.5	0.3

Let us consider the design of a PID compensator using frequency response techniques for a system with a time delay so that

$$G(s) = \frac{Ke^{-Ts}}{(\tau s + 1)}. \quad (12.48)$$

This type of plant represents many industrial processes that incorporate a time delay. We use a PID compensator to introduce two equal zeros so that

$$G_c(s) = \frac{K_2(\tau_1 s + 1)^2}{s}. \quad (12.49)$$

The design method is as follows:

1. Plot the uncompensated Bode diagram for $K_2 G(s)/s$ with a gain K_2 that satisfies the steady-state error requirement.
2. Place the two equal zeros at or near the crossover frequency, ω_c .
3. Test the results and adjust K_2 or the zero locations, if necessary.

EXAMPLE 12.10 PID control of a system with a delay

Consider the system of Fig. 12.16 when

$$G(s) = \frac{Ke^{-0.1s}}{(0.1s + 1)}, \quad (12.50)$$

where $K = 20$ is selected to achieve a small steady-state error for a step input, and where $G_p(s) = 1$. We want an overshoot to a step input of less than 5%.

Plotting the Bode diagram for $G(j\omega)$, we find that the uncompensated system has a negative phase margin and that the system is unstable.

We will use a PID controller of the form of Eq. (12.49) to attain a desirable phase margin of 70° . Then the loop transfer function is

$$GG_c(s) = \frac{20e^{-0.1s}(\tau_1 s + 1)^2}{s(0.1s + 1)}, \quad (12.51)$$

where $K_2 K = 20$. We plot the Bode diagram without the two zeros, as shown in Fig. 12.18. The phase margin is -32° , and the system is unstable prior to the introduction of the zeros.

Since we have introduced a pole at the origin because of the internal term in the PID compensator, we may reduce the gain $K_2 K$ because e_{ss} is now zero. We place the two zeros at or near the crossover, $\omega_c = 11$. We choose to set $\tau_1 = 0.06$ so that the two zeros are set

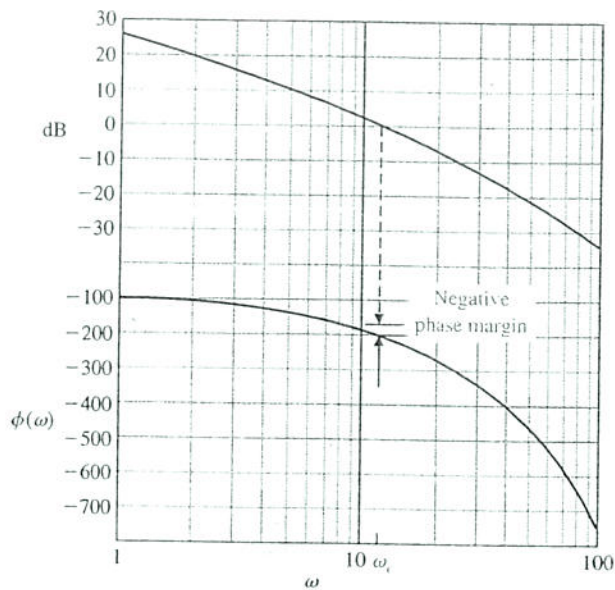


FIGURE 12.18
Bode diagram
for $G(s)/s$ for
Example 12.10.

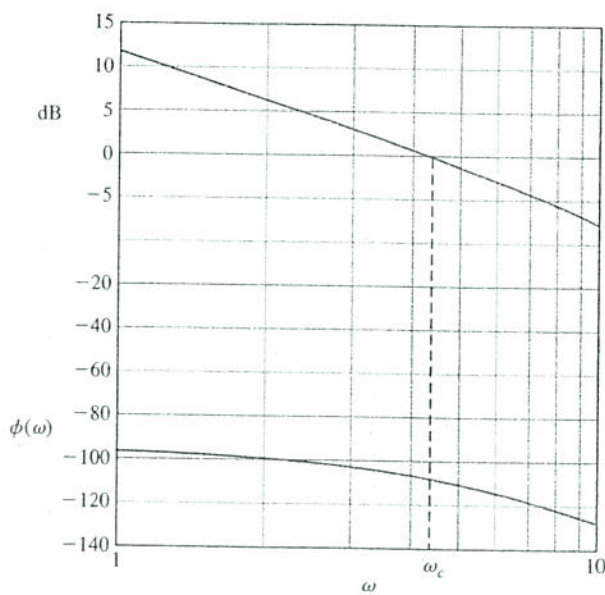


FIGURE 12.19
Bode diagram
for $G_c G(s)$ for
Example 12.10.

at $\omega = 16.7$. Also, we reduce the gain to $K_2 K = 4.5$. Then we obtain the frequency response shown in Fig. 12.19, where

$$G_c G(s) = \frac{4.5(0.06s + 1)^2 e^{-0.1s}}{s(0.1s + 1)} \quad (12.52)$$

The new crossover frequency is $\omega_c = 4.5$, and the phase margin is 70° . The step response of this system has no overshoot and has a settling time (2% criterion) of 0.80 second. This response satisfies the requirements. However, if we wanted to adjust the system further, we could raise K_2K to 10 and achieve a somewhat faster response with an overshoot of less than 5%. ■

As a final consideration of the design of robust control systems using a PID controller, we turn to an s -plane root locus method. This design approach may be simply stated as follows:

1. Place the poles and zeros of $G(s)/s$ on the s -plane.
2. Select a location for the zeros of $G_c(s)$ that will result in an acceptable root locus and suitable dominant roots.
3. Test the transient response of the compensated system and iterate step 2, if necessary.

The root locus design method is illustrated by the following design example for an aircraft autopilot.

12.8 DESIGN EXAMPLE: AIRCRAFT AUTOPILOT

A typical aircraft autopilot control system consists of electrical, mechanical, and hydraulic devices that move the flaps, elevators, fuel-flow controllers, and other components that cause the aircraft to vary its flight. Sensors provide information on velocity, heading, rate of rotation, and other flight data. This information is combined with the desired flight characteristics (commands) electronically available to the autopilot. The autopilot should be able to fly the aircraft on a heading and under conditions set by the pilot. The command often consists of a predetermined heading. Design often focuses on a forward-moving aircraft that moves somewhat up or down without moving right or left and without rolling (rotating the wingtips). Such a study is called pitch axis design. The aircraft is represented by a plant [26]

$$G(s) = \frac{K}{(s + 1/\tau)(s^2 + 2\zeta_1\omega_1s + \omega_1^2)}, \quad (12.53)$$

where τ is the time constant of the actuator. Let $\tau = 1/4$, $\omega_1 = 2$, and $\zeta = 1/2$. Then the s -plane plot has two complex poles—a pole at the origin and a pole at $s = -4$, as shown in Fig. 12.20. The complex poles, representing the aircraft dynamics, can vary within the dashed-line box shown in the figure. We then choose the zeros of the controller as $s = -1.3 \pm j2$, as shown. We select the gain K so that the roots r_2 and $\hat{r}_2$ are complex with a ζ of $1/\sqrt{2}$. The other roots r_1 and $\hat{r}_1$ lie very near the zeros. Therefore the closed-loop transfer function is approximately

$$T(s) \cong \frac{\omega_n^2}{s^2 + 2\zeta\omega_ns + \omega_n^2} = \frac{5}{s^2 + 3.16s + 5}, \quad (12.54)$$

since $\omega_n = \sqrt{5}$, and $\zeta = 1/\sqrt{2}$. The resulting response to a step input has an overshoot of 4.5% and a settling time (2% criterion) of 2.5 seconds, as expected.

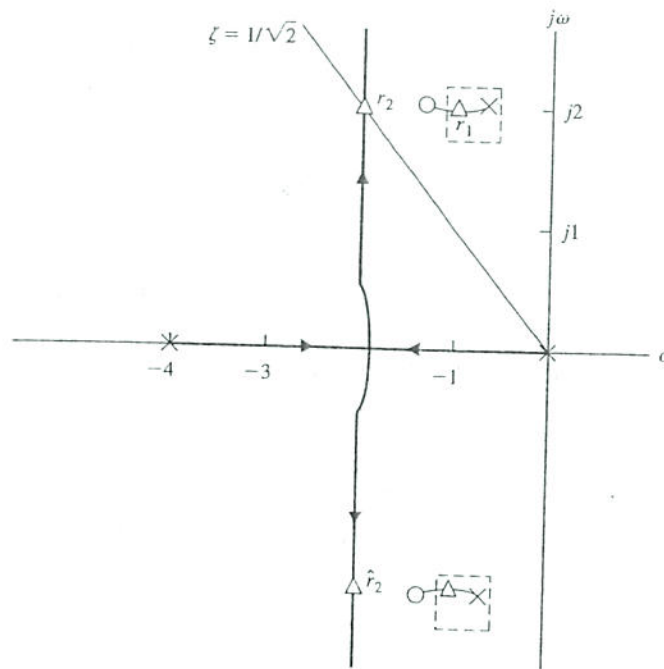


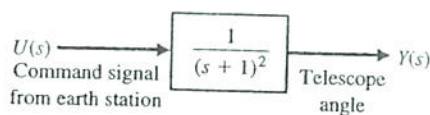
FIGURE 12.20
Root locus for
aircraft autopilot.
The complex poles
can vary within the
dashed-line box
(in color).

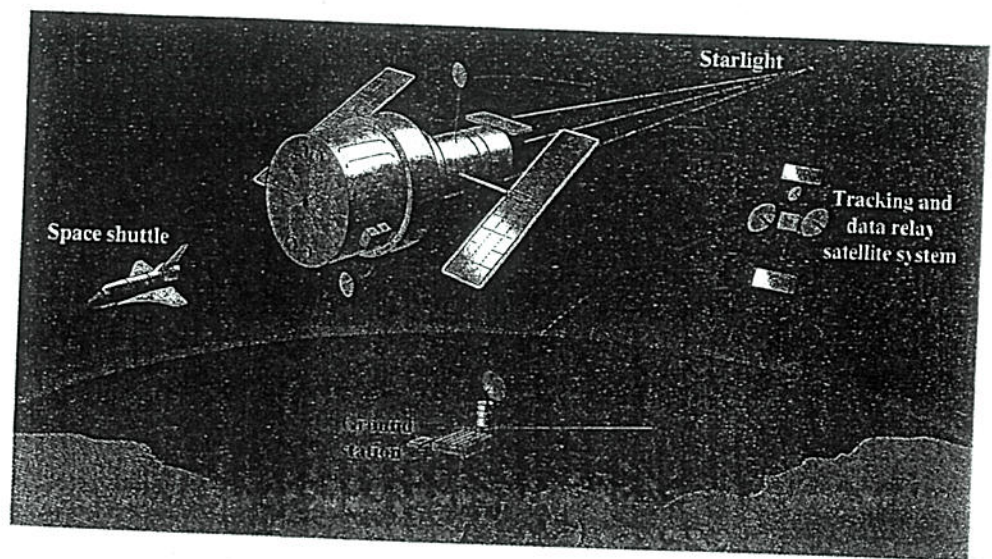
12.9 THE DESIGN OF A SPACE TELESCOPE CONTROL SYSTEM

Scientists have proposed the operation of a space vehicle as a space-based research laboratory and test bed for equipment to be used on a manned space station. The industrial space facility (ISF) would remain in space, and the astronauts would be able to use it only when the shuttle is attached [16, 21]. The ISF will be the first permanent, human-operated commercial space facility designed for R&D, testing, and, eventually, processing in the space environment.

We will consider an experiment operated in space but controlled from Earth. The goal is to manipulate and position a small telescope to accurately point at a planet. The goal is to have a steady-state error equal to zero, while maintaining a fast response to a step with an overshoot of less than 5%. The actuator chosen is a low-power actuator, and the model of the combined actuator and telescope is shown in Fig. 12.21. The command signal is received from an Earth station with a delay of $\pi/16$ seconds. A sensor will measure the pointing direction of the telescope accurately. However, this measurement is relayed back

FIGURE 12.21
Model of a low-
power actuator
and telescope.





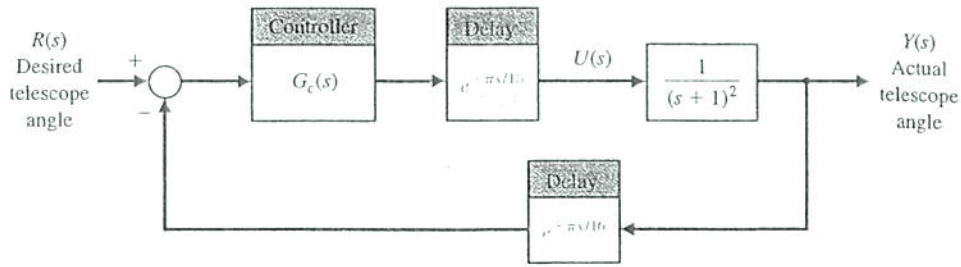


FIGURE 12.22
Feedback control
system for the
telescope
experiment.

to Earth with a delay of $\pi/16$ seconds. Thus the total transfer function of the telescope, actuator, sensor, and round-trip delay (Fig. 12.22) is

$$G(s) = \frac{e^{-s\pi/8}}{(s+1)^2}. \quad (12.55)$$

We propose a controller that is the three-term controller described in Section 12.6, where

$$G_c(s) = K_1 + \frac{K_2}{s} + K_3s = \frac{K_1s + K_2 + K_3s^2}{s}. \quad (12.56)$$

The use of only the proportional term will not be acceptable since we require a steady-state error of zero for a step input. Thus we must use a finite value of K_2 , and hence we may elect to use either proportional plus integral control (PI) or proportional plus integral plus derivative control (PID).

We will first try PI control, so that

$$G_c(s) = K_1 + \frac{K_2}{s} = \frac{K_1s + K_2}{s}. \quad (12.57)$$

Since we have a pure delay, e^{-sT} , we use the frequency response methods for the design process. Thus we will translate the overshoot specification to the frequency domain. If we have two dominant characteristic roots, the overshoot to a step is 5% when $\zeta = 0.7$, or the phase margin requirement is about 70° .

If we choose $K_1 = 0.022$ and $K_2 = 0.22$, we have

$$G_c(s)G(s) = \frac{0.22(0.1s + 1)e^{-s\pi/8}}{s(s+1)^2}, \quad (12.58)$$

and the Bode diagram is shown in Fig. 12.23. The location of the zero at $s = 10$ was chosen to add a phase lead angle in order to attain the desired phase margin. An iterative procedure yields a series of trials for K_1 and K_2 until the desired phase margin is achieved. Note that we have achieved a phase margin of about 63° . The actual step response was plotted, and it is determined that the overshoot is 4.7% with a settling time (2% criterion) of 16 seconds, as recorded in Table 12.5.

The proportional plus integral plus derivative controller is

$$G_c(s) = \frac{K_1s + K_2 + K_3s^2}{s}. \quad (12.59)$$

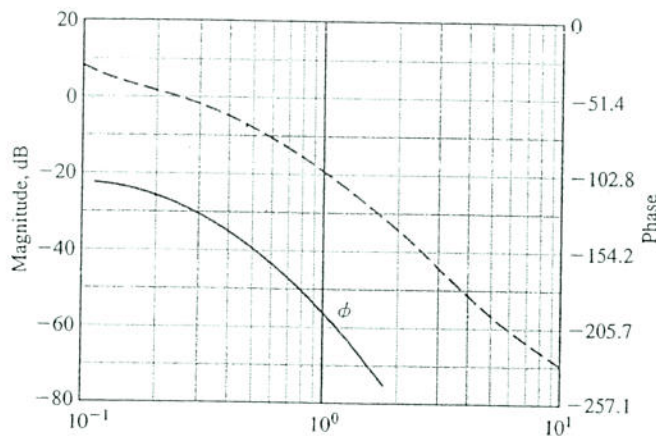


FIGURE 12.23
Bode diagram for
the system with the
PI controller.

TABLE 12.5 Step Response of the Space Telescope for Two Controllers

	Steady-State Error	Percent Overshoot	Settling Time (seconds)
PI controller	0	4.7	16.0
PID controller	0	3.7	5.8

We now have three parameters to vary to achieve the desired phase margin. If we select, after some iteration, $K_1 = 0.8$, $K_2 = 0.5$, and $K_3 = 10^{-3}$, we obtain a phase margin of 64° . The percentage overshoot is 3.7%, and the settling time (2% criterion) is 5.8 seconds. Perhaps the easiest way to select the gain constants is to let K_3 be a small, but nonzero, number initially and $K_1 = K_2 = 0$, then plot the frequency response. In this case, we choose $K_3 = 10^{-3}$ and obtain a Bode plot. We then use $K_1 \approx K_2$ and iterate to obtain the appropriate values of these unspecified gains.

The performance of the PI- and the PID-compensated systems is recorded in Table 12.5. The PID controller is the most desirable, since it provides a shorter settling time.

12.10 THE DESIGN OF A ROBUST BOBBIN DRIVE

Monofilament nylon is produced by an extrusion process that outputs filament at a constant rate. The product is wound onto a bobbin that rotates at a maximum speed of 2000 rpm. The tension in the filament must be held between 0.2 and 0.6 pound to ensure that it is not stretched. The winding diameter varies between 2 to 4 inches.

The filament is laid onto the bobbin by a ballscrew-driven arm that oscillates back and forth at constant speed, as shown in Fig. 12.24(a). The arm must reverse rapidly at the end of the move. The required ballscrew speed is 60 rpm. The prime requirement of the bobbin

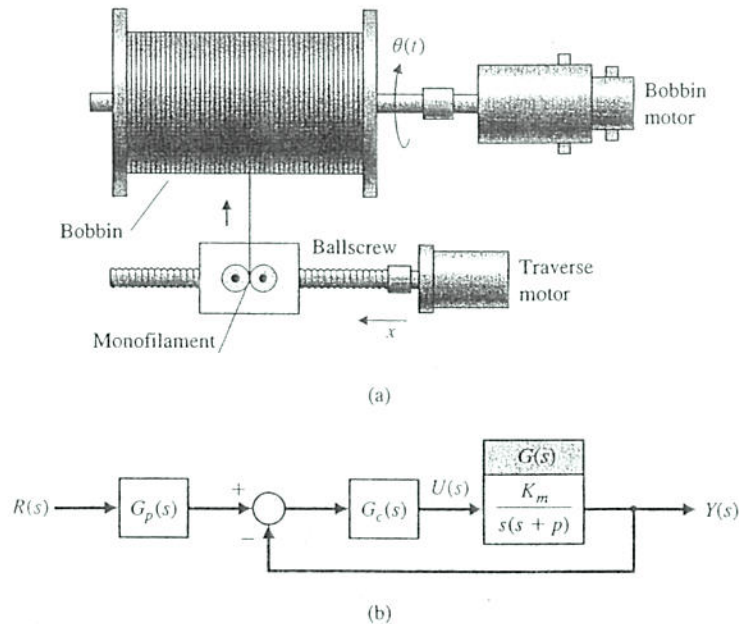


FIGURE 12.24
A monofilament
bobbin winder.

drive is to provide a controlled tension. Since the winding diameter varies by 2 to 1, the tension will fall by 50% from start to finish.

The control system will have a system structure as shown in Fig. 12.24(b), for which we select a PID controller. The parameter variations are $1.5 \leq K_m \leq 2.5$, and $3 \leq p \leq 5$ with the nominal conditions $K_m = 2$, and $p = 4$. Furthermore a third pole at $s = -50$ has been omitted from the model. The requirements are an overshoot less than 2.5% and a settling time (2% criterion) less than 0.4 second. The magnitude of $u(t)$ must be less than 100.

Using a PID controller, the ITAE design, and the nominal parameters, we determine ω_n from the settling time requirement. Since we expect that $\zeta \approx 0.8$, we use

$$T_s = \frac{4}{0.8\omega_n} < 0.4.$$

We select $\omega_n = 23$ as the maximum allowable for $|u| < 100$. Then for

$$G_c(s) = K_1 + \frac{K_2}{s} + K_3s,$$

we obtain $K_1 = 568.68$, $K_2 = 6083.5$, and $K_3 = 18.13$. Using the appropriate prefilter, the response is as recorded in Table 12.6. The system does not offer robust performance since the overshoot requirement is not satisfied when the worst-case parameters are considered.

We also examine the performance of the system with the nominal parameters but with the unmodeled pole added, so that the actual plant is

$$G(s) = \frac{2(50)}{s(s+4)(s+50)}. \quad (12.60)$$

TABLE 12.6 Response of the Bobbin Drive System for a Unit Step Input (original design)

	Parameters	Percent Overshoot	Settling Time	$\left \frac{u(t)}{r(t)} \right _{\text{maximum}}$
Nominal parameters	$K_m = 2,$ $p = 4$	1.96%	0.318	98
Worst-case parameters	$K_m = 1.5,$ $p = 3$	7.48%	0.375	95
Nominal parameters and added third pole at $s = -50$		9.82%	0.732	90

(Note that the dc gain, $\lim_{s \rightarrow 0} sG(s)$, remains 0.5.) The response of the PID controller with the added pole is recorded in Table 12.6. Again the system fails the requirement of robust performance.

We need to adjust the system so that the performance with the worst-case parameters is acceptable. Examine the root locus for the nominal parameters shown in Fig. 12.25. Insert a cascade gain K prior to $G_c(s)$ so that we have $KG_c(s)$. Then the roots for $K = 1$ and $K = 3$ are shown on the locus. Since the worst-case response occurs when the motor constant K_m drops to 1.5, we use the cascade gain $K = 3$ to move the roots to the left on

FIGURE 12.25 Root locus for the normal case and the worst case for $K = 1$ and $K = 3$.

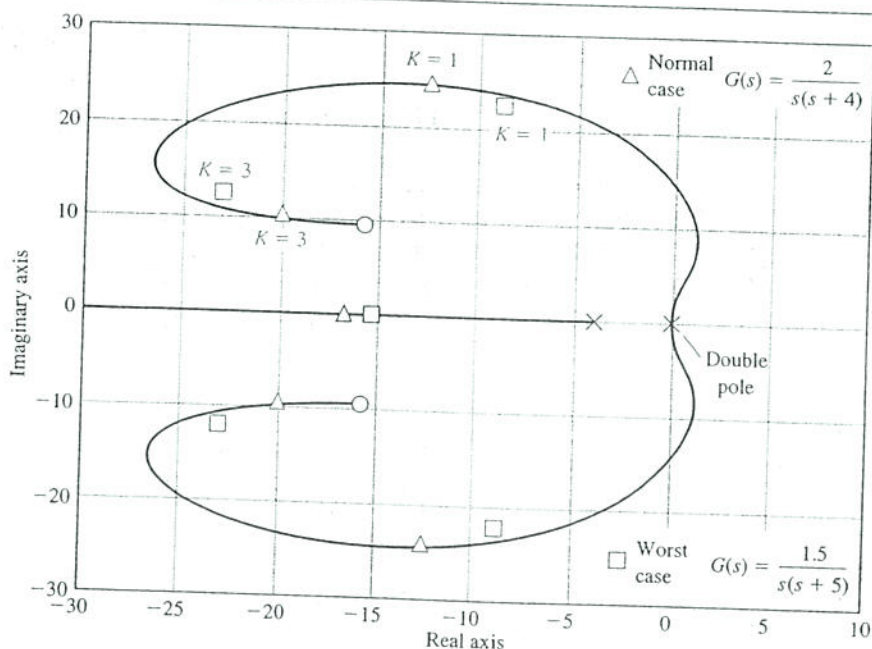


TABLE 12.7 Response of the Bobbin Drive System for a Unit Step with Additional Cascade Gain $K = 3$

	Percent Overshoot	Settling Time (seconds)
Nominal parameters	0.12%	0.218
Worst-case parameters	0.47%	0.214
Nominal parameters and third pole	0.50%	0.242

the s -plane. Then when the gain K_m drops to 1.5, the roots still are in the desired region. The response of the system with $K = 3$ is recorded in Table 12.7 for the nominal and worst-case conditions, and with the added pole. This system meets all the specifications. This approach uses a cascade gain that, when adjusted correctly, will drive the dominant roots near the complex zeros of the PID controller. Then when the worst parameter change occurs, the system will still maintain the required performance.

E.

12.11 THE ROBUST INTERNAL MODEL CONTROL SYSTEM

The internal model control system is shown in Fig. 12.26 and was previously considered in Section 11.8. We now reconsider the use of the internal model design with special attention to robust system performance. The **internal model principle** states that if $G_c(s)G(s)$ contains $R(s)$, then $y(t)$ will track $r(t)$ asymptotically (in the steady state), and the tracking is robust.

Examining the system of Fig. 12.26, we note that for lower-order plants, state variable feedback will not be required, and a suitable $G_c(s)$ can be obtained. However, with higher-order systems, the feedback of all state variables may be required.

Consider a simple system with $G(s) = 1/s$, for which we seek a ramp response with a

FIGURE 12.26
The internal model control system.

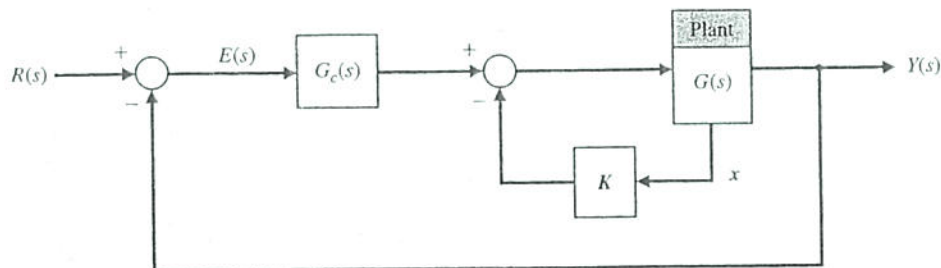


FIG
An
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steady-state error of zero. A PI controller is sufficient, and we let $\mathbf{K} = \mathbf{0}$ (no state variable feedback). Then we have

$$G_c G(s) = \left(K_1 + \frac{K_2}{s} \right) \frac{1}{s} = \frac{(K_1 s + K_2)}{s^2}. \quad (12.61)$$

Note that for a ramp, $R(s) = 1/s^2$, which is contained as a factor of Eq. (12.61), and the closed-loop transfer function is

$$T(s) = \frac{(K_1 s + K_2)}{s^2 + K_1 s + K_2}. \quad (12.62)$$

Using the ITAE specifications for a ramp response (Table 5.7), we require

$$T(s) = \frac{3.2\omega_n s + \omega_n^2}{s^2 + 3.2\omega_n s + \omega_n^2}. \quad (12.63)$$

We select ω_n to satisfy a specification for the settling time. For a settling time (2% criterion) of 1 second, we select $\omega_n = 5$. Then we require $K_1 = 16$ and $K_2 = 25$. The response of this system settles in 1 second and then tracks the ramp with zero steady-state error. If this system (designed for a ramp input) receives a step input, the response has an overshoot of 5% and a settling time of 1.5 seconds. This system is very robust to changes in the plant. For example, if $G(s) = K/s$ changes gain so that K shifts from $K = 1$ by $\pm 50\%$, the change in the ramp response is insignificant.

EXAMPLE 12.11 Design of an internal model control system

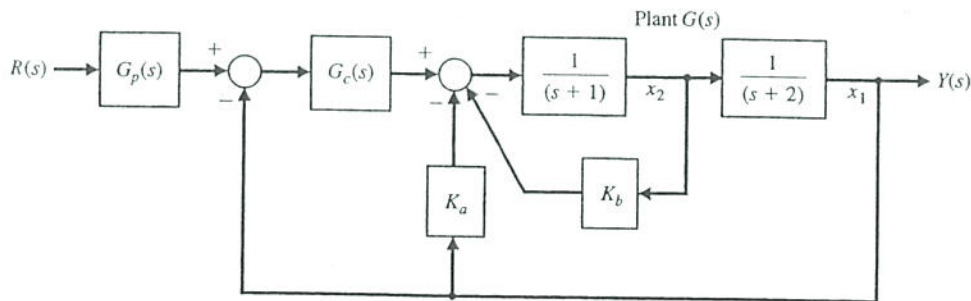
Consider the system of Fig. 12.27 with state variable feedback and a compensator $G_c(s)$. We wish to track a step input with zero steady-state error. Here we select a PID controller for $G_c(s)$. We then have

$$G_c(s) = \frac{K_3 s^2 + K_1 s + K_2}{s},$$

and $G(s)G_c(s)$ will contain $R(s) = 1/s$, the input command. Note that we feed back both state variables and add these additional signals after $G_c(s)$ in order to retain the integrator in $G_c(s)$.

The goal is to achieve a settling time to within 2% of the final value in less than 1 second and a deadbeat response (see Section 10.11) while retaining a robust response. Here

FIGURE 12.27
An internal model control with state variable feedback and $G_c(s)$.



we assume that the two poles of $G(s)$ can change by $\pm 50\%$. Then the worst-case condition is

$$\hat{G}(s) = \frac{1}{(s + 0.5)(s + 1)}.$$

One design approach is to design the control for this worst-case condition. Another approach, which we use here, is to design for the nominal $G(s)$ and one-half the desired settling time. Then we expect to meet the settling time requirement and attain a very fast, highly robust system. Note that the prefilter $G_p(s)$ is used to attain the desired form for $T(s)$.

The response desired is deadbeat (see Table 10.2), so use a third-order transfer function as

$$T(s) = \frac{\omega_n^3}{s^3 + 1.9\omega_n s^2 + 2.20\omega_n^2 s + \omega_n^3}, \quad (12.64)$$

and the settling time (2% criterion) is $T_s = 4.04/\omega_n$. For a settling time of $1/2$ second, we use $\omega_n = 8.08$.

The closed-loop transfer function of the system of Fig. 12.27 with the appropriate $G_p(s)$ is

$$T(s) = \frac{K_2}{s^3 + (3 + K_3 + K_b)s^2 + (2 + K_1 + K_a + 2K_b)s + K_2}. \quad (12.65)$$

We let $K_a = 10$, $K_b = 2$, $K_1 = 127.6$, $K_2 = 527.5$, and $K_3 = 10.35$. Note that $T(s)$ could be achieved with other gains, including $K_b = 0$.

The step response of this system has a deadbeat response with an overshoot of 1.65% and a settling time of 0.5 second. When the plant poles of $G(s)$ change by $\pm 50\%$, the overshoot changes to 1.86%, and the settling time becomes 0.95 second. This is an outstanding design of a very robust, deadbeat response system. ■

12.12 THE DESIGN OF AN ULTRA-PRECISION DIAMOND TURNING MACHINE

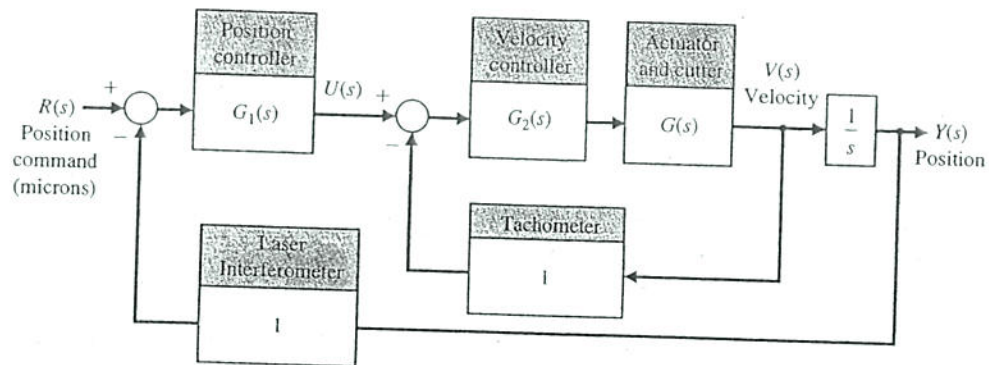
The design of an ultra-precision diamond turning machine has been studied at Lawrence Livermore National Laboratory. This machine shapes optical devices such as mirrors with ultra-high precision using a diamond tool as the cutting device. In this discussion we will consider only the z -axis control. Using frequency response identification with sinusoidal input to the actuator, it was determined that

$$G(s) = \frac{4500}{(s + 60)}. \quad (12.66)$$

The system can accommodate high gains, such as 4500, since the input command, $r(t)$, is a series of step commands of very small magnitude (a fraction of a micron). The system has an outer loop for position feedback using a laser interferometer with an accuracy of 0.1 micron (10^{-7} m). An inner feedback loop is also used for velocity feedback, as shown in Fig. 12.28.

It is desired to select the controllers, $G_1(s)$ and $G_2(s)$, to obtain an overdamped, highly robust, high-bandwidth system. The robust system must accommodate changes in $G(s)$ due to varying loads, materials, and cutting requirements. Thus we seek large phase margin and

FIGURE 12.28
Turning machine
control system.



gain margin for the inner and outer loops, and low root sensitivity. The specifications are summarized in Table 12.8.

Since we desire zero steady-state error for the velocity loop, we use a velocity loop controller $G_2(s) = G_3(s)G_4(s)$, where $G_3(s)$ is a PI controller, and $G_4(s)$ is a lead controller. We utilize

$$G_2(s) = G_3(s)G_4(s) = \frac{(1 + K_3 s)}{K_3 s} \cdot \frac{(1 + K_4 s)}{\alpha \left(1 + \frac{K_4}{\alpha} s\right)} \cdot K_2$$

and choose $K_3 = 0.00532$, $K_4 = 0.00272$, and $\alpha = 2.95$. We now have

$$G_2(s) = K_2 \frac{(s + 188)}{s} \cdot \frac{(s + 368)}{(s + 1085)}$$

TABLE 12.8 Specifications for Turning Machine Control System

Specification	Transfer Function	
	Velocity, $V(s)/U(s)$	Position, $Y(s)/R(s)$
Minimum bandwidth	950 rad/s	95 rad/s
Steady-state error to a step	0	0
Minimum damping ratio, ζ	0.8	0.9
Maximum root sensitivity, $ S_K^r $	1.0	1.5
Minimum phase margin	90°	75°
Minimum gain margin	40 dB	60 dB

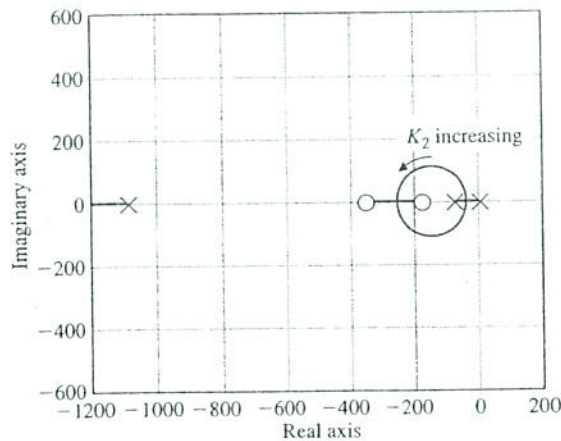


FIGURE 12.29
Root locus for
velocity loop as
 K_2 varies.

The root locus for $G_2(s)G(s)$ is shown in Fig. 12.29. When $K_2 = 2$, we have for the velocity closed-loop transfer function

$$T_2(s) = \frac{V(s)}{U(s)} = \frac{9000(s + 188)(s + 368)}{(s + 205)(s + 305)(s + 10^4)} \approx \frac{10^4}{(s + 10^4)}, \quad (12.67)$$

which is a large-bandwidth system. The actual bandwidth and root sensitivity are summarized in Table 12.9. Note that we have exceeded the specifications for the velocity transfer function.

We will use a lead network for the position loop as

$$G_1(s) = K_1 \frac{(1 + K_5 s)}{\alpha \left(1 + \frac{K_5}{\alpha} s\right)}$$

TABLE 12.9 Design Results for Turning Machine Control System

Achieved Result	Velocity Transfer Function	Position Transfer Function, $Y(s)/R(s)$
Closed-loop bandwidth	4000 rad/s	1000 rad/s
Steady-state error	0	0
Damping ratio, ζ	1.0	1.0
Root sensitivity, $ S'_K $	0.92	1.2
Phase margin	93°	85°
Gain margin	Infinite	76 dB

and we choose $\alpha = 2.0$ and $K_5 = 0.0185$ so that

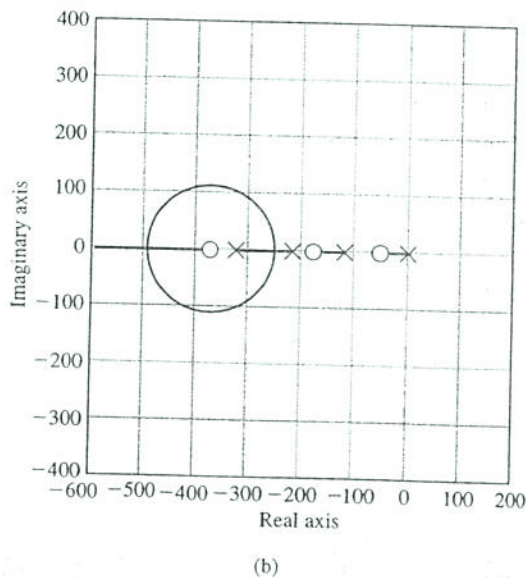
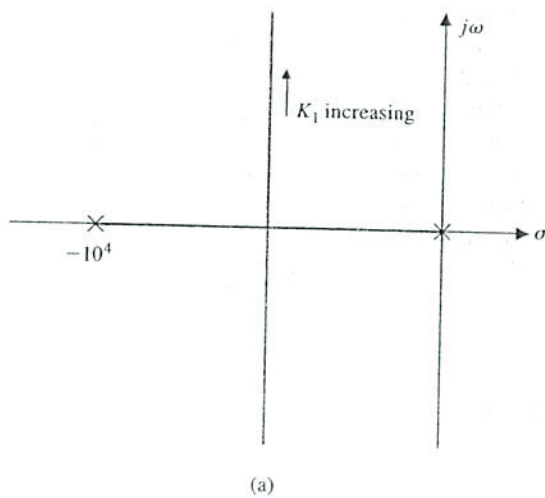
$$G_1(s) = \frac{K_1(s + 54)}{(s + 108)}.$$

We then plot the root locus for

$$G_1(s) \cdot T_2(s) \cdot \frac{1}{s}.$$

If we use the approximate $T_2(s)$ of Eq. (12.73), we have the root locus of Fig. 12.30(a). Using the actual $T_2(s)$, we get the close-up of the root locus shown in Fig. 12.30(b). We select $K_2 = 1000$ and achieve the actual results for the total system transfer function as

FIGURE 12.30
The root locus
for $K_1 > 0$ for
(a) overview and
(b) close-up near
origin of the
s-plane.



recorded in Table 12.9. The total system has a high phase margin, a low sensitivity, and is overdamped with a large bandwidth. This system is very robust.

12.13 THE PSEUDO-QUANTITATIVE FEEDBACK SYSTEM

Quantitative feedback theory (QFT) uses a controller, as shown in Fig. 12.31, to achieve robust performance. The goal is to achieve a wide bandwidth for the closed-loop transfer function with a high loop gain K . Typical QFT design methods use graphical and numerical methods in conjunction with the Nichols chart. Generally QFT design seeks high loop gain and large phase margin so that robust performance is achieved [27–29, 32].

In this section we pursue a simple method of achieving the goals of QFT with an s -plane, root locus approach to the selection of the gain K and the compensator $G_c(s)$. This approach, dubbed pseudo-QFT, follows these steps:

1. Place the n poles and m zeros of $G(s)$ on the s -plane for the n th order $G(s)$. Also add any poles of $G_c(s)$.
2. Starting near the origin, place the zeros of $G_c(s)$ immediately to the left of each of the $(n - 1)$ poles on the left-hand s -plane. This leaves one pole far to the left of the left-hand side of the s -plane.
3. Increase the gain K so that the roots of the characteristic equation (poles of the closed-loop transfer function) are close to the zeros of $G_c G(s)$.

This method introduces zeros so that all but one of the root loci end on finite zeros. If the gain K is sufficiently large, then the poles of $T(s)$ are almost equal to the zeros of $G_c G(s)$. This leaves one pole of $T(s)$ with a significant partial fraction residue and the system with a phase margin of approximately 90° (actually about 85°).

EXAMPLE 12.12 Design using the pseudo-QFT method

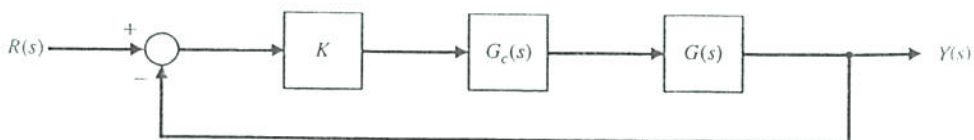
Consider the system of Fig. 12.31 with

$$G(s) = \frac{1}{(s + p_1)(s + p_2)},$$

where the nominal case is $p_1 = 1$ and $p_2 = 2$, with $\pm 50\%$ variation. The worst case is with $p_1 = 0.5$ and $p_2 = 1$. We wish to design the system for zero steady-state error for a step input, so we use the PID controller

$$G_c(s) = \frac{(s + z_1)(s + z_2)}{s}.$$

FIGURE 12.31
Feedback system.



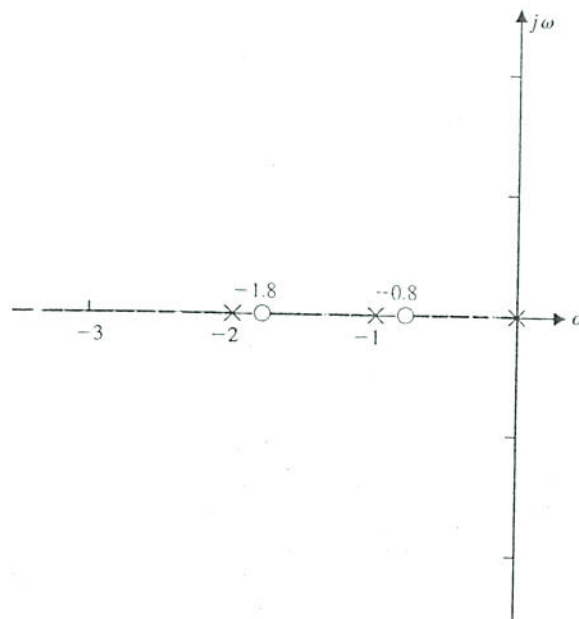


FIGURE 12.32
Root locus for
 $KG_c G(s)$.

We then invoke the internal model principle, with $R(s) = 1/s$ incorporated within $G_c G(s)$. Using step 1, we place the poles of $G_c G(s)$ on the s -plane, as shown in Fig. 12.32. There are three poles, at $s = 0$, -1 , and -2 , as shown. Step 2 calls for placing a zero to the left of the pole at the origin and at the pole at $s = -1$, as shown in Fig. 12.32.

The compensator is thus

$$G_c(s) = \frac{(s + 0.8)(s + 1.8)}{s} \quad (12.68)$$

We select $K = 100$, so that the roots of the characteristic equation are close to the zeros. The closed-loop transfer function is

$$T(s) = \frac{100(s + 0.80)(s + 1.80)}{(s + 0.798)(s + 1.797)(s + 100.4)} \approx \frac{100}{(s + 100)} \quad (12.69)$$

This closed-loop system provides a fast response and possesses a phase margin of approximately 85° . The performance is summarized in Table 12.10.

TABLE 12.10 Performance of
Pseudo-QFT Design

	Percent Overshoot	Settling Time
Nominal $G(s)$	0.01%	40 ms
Worst-case $G(s)$	0.97%	40 ms

When the worst-case conditions are realized ($p_1 = 0.5$ and $p_2 = 1$), the performance remains essentially unchanged, as shown in Table 12.10. Pseudo-QFT design results in very robust systems. ■

12.14 ROBUST CONTROL SYSTEMS USING MATLAB

In this section we will investigate robust control systems using MATLAB. In particular we will consider the commonly used PID controller in the feedback control system shown in Fig. 12.1. Notice that the system has a **prefilter** $G_p(s)$. The contribution of the prefilter to optimum performance is discussed in Section 10.10.

The PID controller has the form

$$G_c(s) = \frac{K_3 s^2 + K_1 s + K_2}{s}.$$

Notice that the PID controller is not a rational function (i.e., the degree of the numerator polynomial is greater than the degree of the denominator polynomial). One will experience difficulty if the PID controller is input into MATLAB in the standard numerator and denominator fashion. The problem can be resolved by utilizing the `conv` function rather than the `series` function in the manipulations.

The objective is to choose the parameters K_1 , K_2 , and K_3 to meet the performance specifications and have desirable robustness properties. Unfortunately it is not immediately clear how to choose the parameters in the PID controller to obtain certain robustness characteristics. An illustrative example will show that it is possible to choose the parameters iteratively and verify the robustness by simulation. Using MATLAB helps in this process, since the entire design and simulation can be mechanized utilizing scripts and can easily be executed repeatedly.

EXAMPLE 12.13 Robust control of temperature

Consider the feedback control system in Fig. 12.1, where

$$G(s) = \frac{1}{(s + c_0)^2},$$

the nominal value is $c_0 = 1$, and $G_p(s) = 1$. We will design a compensator based on $c_0 = 1$ and check robustness by simulation. Our design specifications are as follows:

1. settling time (2% criterion) $T_s \leq 0.5$ second, and
2. optimum ITAE performance for a step input.

For this design we will not utilize a prefilter to meet specification (2) but will instead show that acceptable performance (i.e., low overshoot) can be obtained by increasing a cascade gain.

The closed-loop transfer function is

$$T(s) = \frac{K_3 s^2 + K_1 s + K_2}{s^3 + (2 + K_3)s^2 + (1 + K_1)s + K_2}. \quad (12.70)$$

The associated root locus equation is

$$1 + \hat{K} \left(\frac{s^2 + as + b}{s^3} \right) = 0,$$

where

$$\hat{K} = K_3 + 2, \quad a = \frac{1 + K_1}{2 + K_3}, \quad b = \frac{K_2}{2 + K_3}.$$

The settling time requirement $T_s < 0.5$ second leads us to choose the roots of $(s^2 + as + b)$ to the left of the $s = -\zeta\omega_n = -8$ line in the s -plane, as shown in Fig. 12.33, to ensure that the locus travels into the required s -plane region. We have chosen $a = 16$ and $b = 70$ to ensure the locus travels past the $s = -8$ line. We select a point on the root locus in the performance region, and using the `rlocfind` function, we find the associated gain $\hat{K}$ and the associated value of ω_n . For our chosen point we find that

$$\hat{K} = 118.$$

Then with $\hat{K}$, a , and b , we can solve for the PID coefficients as follows:

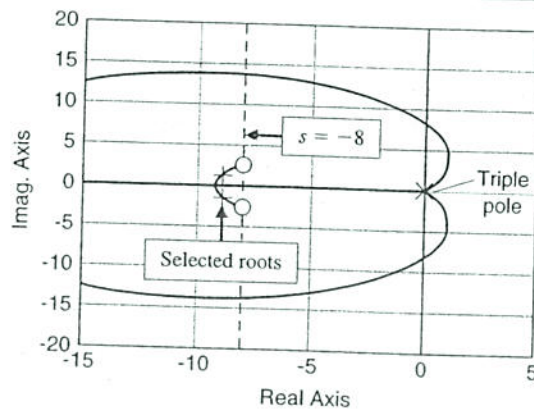
$$K_3 = \hat{K} - 2 = 116,$$

$$K_1 = a(2 + K_3) - 1 = 1187,$$

$$K_2 = b(2 + K_3) = 8260.$$

To meet the overshoot performance requirements for a step input, we will utilize a cascade gain K that will be chosen by iterative methods using the `step` function, as illustrated in

FIGURE 12.33
Root locus for the
PID-compensated
temperature
controller as
 $\hat{K}$ varies.



```
>>a=16; b=70; num=[1 a b]; den=[1 0 0 0]; rlocus(num,den)
>>rlocfind(num,den)
```

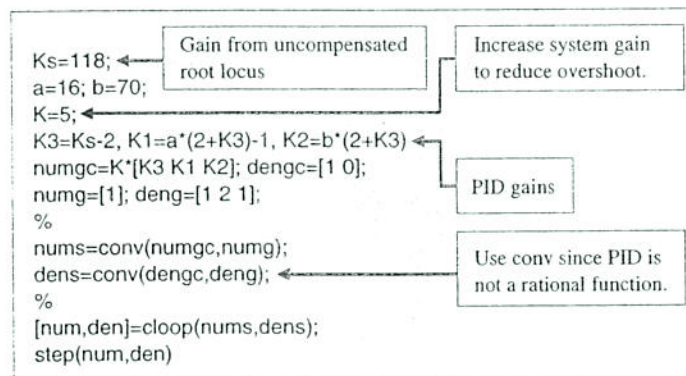
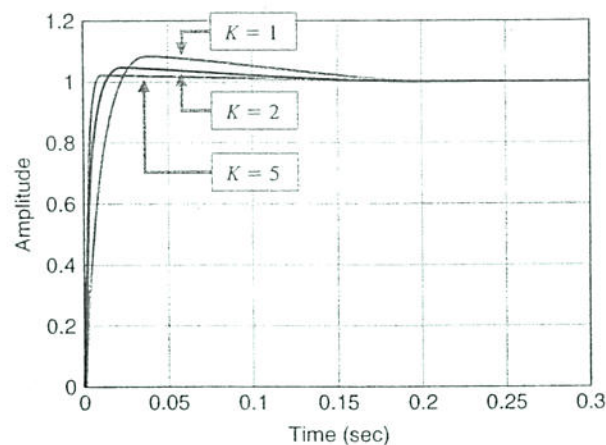


FIGURE 12.34
Step response
of the PID
temperature
controller.

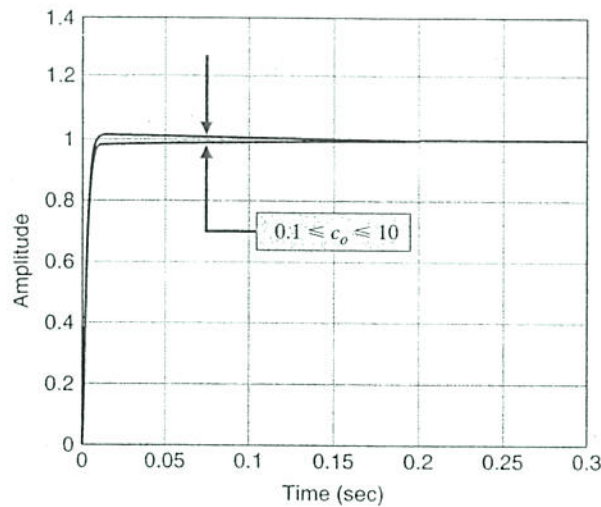
Fig. 12.34. The step response corresponding to $K = 5$ has an acceptable overshoot of 2%. With the addition of the gain $K = 5$, the final PID controller is

$$G_c(s) = K \frac{(K_3 s^2 + K_1 s + K_2)}{s} = 5 \frac{(116 s^2 + 1187 s + 8260)}{s}. \quad (12.71)$$

We do not use the prefilter, as in Example 12.8. Instead we increase the cascade gain K to obtain satisfactory transient response. Now we can consider the question of robustness to changes in the plant parameter c_0 .

The investigation into the robustness of the design consists of a step response analysis using the PID controller given in Eq. (12.71) for a range of plant parameter variations of $0.1 \leq c_0 \leq 10$. The results are displayed in Fig. 12.35. The script is written to compute the step response for a given c_0 . It might be a good idea to place the input of c_0 at the command prompt level to make the script more interactive.

The simulation results indicate that the PID design is robust with respect to changes in c_0 . The differences in the step responses for $0.1 \leq c_0 \leq 10$ are barely discernible on the plot. If the results showed otherwise, it would be possible to iterate on the design until an



```

c0=10 ← Specify plant parameter.
numg=[1]; deng=[1 2*c0 c0^2];
numgc=5*[116 1187 8260]; dengc=[1 0];
%
numa=conv(numg,numgc); dena=conv(deng,dengc);
%
[num,den]=cloop(numa,dena);
%
step(num,den)

```

FIGURE 12.35
Robust PID
controller analysis
with variations
in c_0 .

acceptable performance was achieved. The interactive capability of MATLAB allows us to check the robustness by simulation. ■

12.15 SEQUENTIAL DESIGN EXAMPLE: DISK DRIVE READ SYSTEM



In this chapter we will design a PID controller to achieve the desired system response. Many actual disk drive head control systems use a PID controller and use a command signal $r(t)$ that utilizes an ideal velocity profile at the maximum allowable velocity until the head arrives near the desired track, when $r(t)$ is switched to a step-type input. Thus we desire zero steady-state error for a ramp (velocity) signal and a step signal. Examining the system shown in Fig. 12.36 we note that the forward path possesses two pure integrations, and we expect zero steady-state error for a velocity input $r(t) = At, t > 0$.

The PID controller is

$$G_c(s) = K_1 + \frac{K_2}{s} + K_3 s = \frac{K_3(s + z_1)(s + \hat{z}_1)}{s}$$

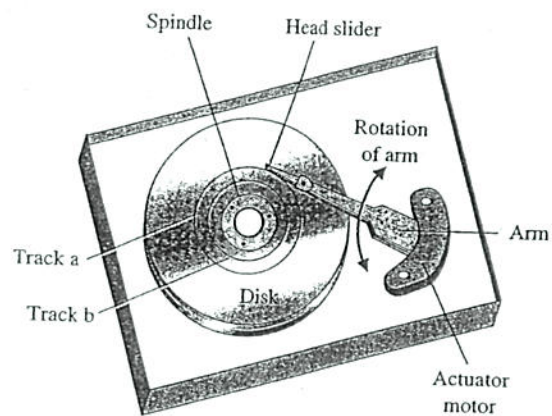


FIGURE 1.24
Diagram of disk drive.

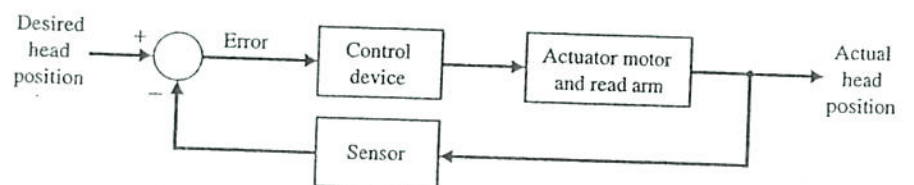


FIGURE 1.25
Closed-loop control system for disk drive.

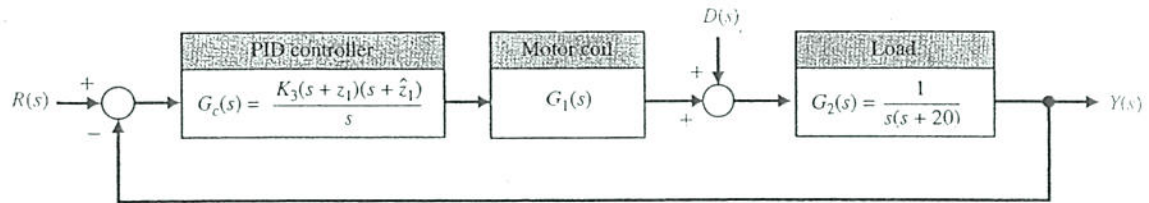


FIGURE 12.36
Disk drive feedback system with a PID controller.

The motor field transfer function is

$$G_1(s) = \frac{5000}{(s + 1000)} \approx 5.$$

The second-order model uses $G_1(s) = 5$, and the design is determined for this model.

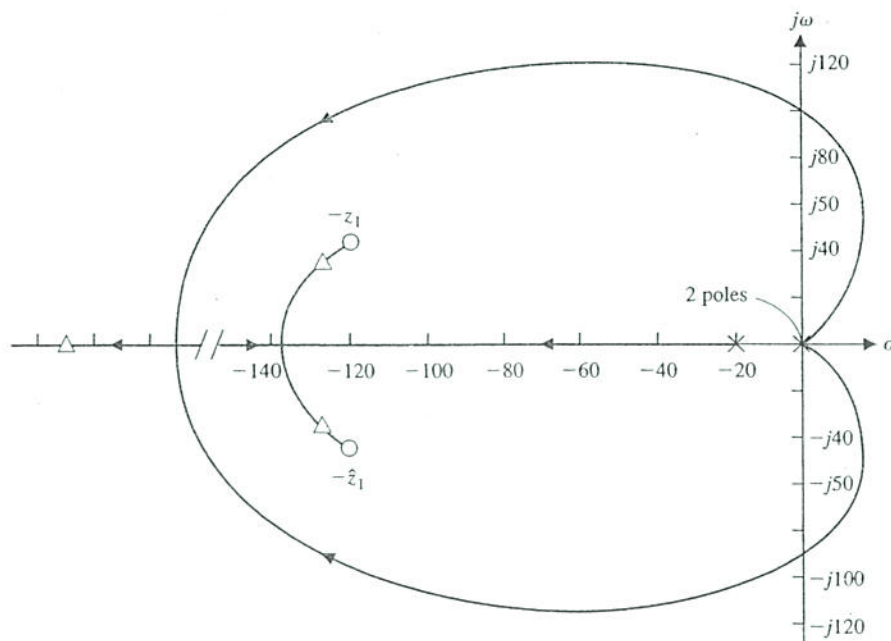
We use the second-order model and the PID controller for the s -plane design technique illustrated in Section 12.6. The poles and zeros of the system are shown in the s -plane in Fig. 12.37 for the second-order model and $G_1(s) = 5$. Then we have

$$G_c G_1 G_2(s) = \frac{5K_3(s + z_1)(s + \hat{z}_1)}{s^2(s + 20)}.$$

We select $z_1 = -120 + j40$ and determine $5K_3$ so that the roots are to the left of the line $s = -100$. If we achieve that requirement, then

$$T_s < \frac{4}{100}.$$

FIGURE 12.37
A sketch of a root locus as K_3 increases for estimated root locations with a desirable system response.



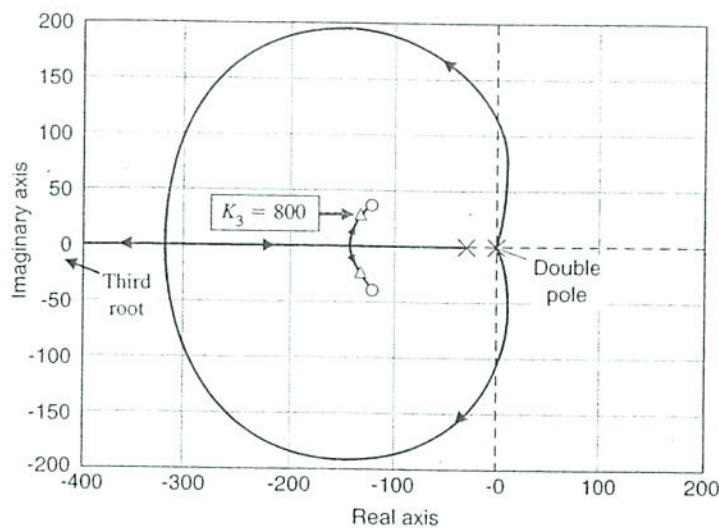


FIGURE 12.38
Actual root locus
for the second-
order model.

TABLE 12.11 Disk Drive Control System
Specifications and Actual Performance

Performance Measure	Desired Value	Response for Second-Order Model
Percent overshoot	$<5\%$	4.5%
Settling time for step input	<50 ms	6 ms
Maximum response for a unit step disturbance	$<5 \times 10^{-3}$	7.7×10^{-7}

and the overshoot to a step input is (ideally) less than 2% since zeta of the complex roots is approximately 0.8. Of course, this sketch is only a first step. As a second step we use MATLAB iteratively to determine K_3 . We then obtain the actual root locus as shown in Fig. 12.38 with $K_3 = 800$. The system response is recorded in Table 12.11. The system meets all the specifications.

12.16 SUMMARY

The design of highly accurate control systems in the presence of significant plant uncertainty requires the designer to seek a robust control system. A robust control system exhibits low sensitivities to parameter change and is stable over a wide range of parameter variations.

The three-mode, or PID, controller was considered as a compensator to aid in the

$$G_c(s) = \frac{V_0(s)}{V_1(s)} = \frac{R_4 R_2 (R_1 C_1 s + 1) (R_2 C_2 s + 1)}{R_3 R_1 (R_2 C_2 s)}$$

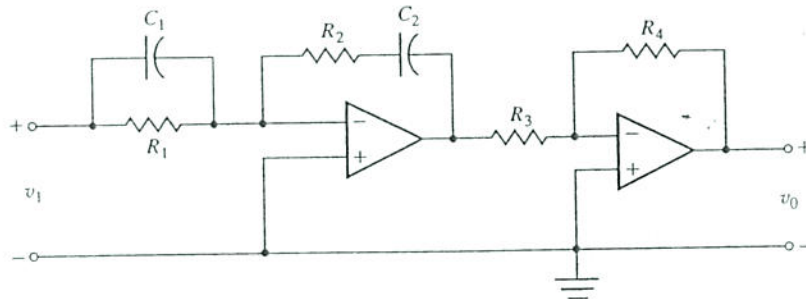


FIGURE 12.39
Operational
amplifier circuit
used for PID
controller.

design of robust control systems. The design issue for a PID controller is the selection of the gain and two zeros of the controller transfer function. We utilized three design methods for the selection of the controller: the root locus method, the frequency response method, and the ITAE performance index method. An operational amplifier circuit used for a PID controller is shown in Fig. 12.39. In general, the use of a PID controller will enable the designer to attain a robust control system.

The internal model control system with state variable feedback and a controller $G_c(s)$ was utilized to obtain a robust control system. Finally the robust nature of a pseudo-QFT control system was demonstrated.

A robust control system provides stable, consistent performance as specified by the designer in spite of wide variation of plant parameters and disturbances. It also provides highly robust response to command inputs and a steady-state tracking error equal to zero.

For systems with uncertain parameters, the need for robust systems will require the incorporation of advanced machine intelligence, as shown in Fig. 12.40.

FIGURE 12.40
Intelligence
required versus
uncertainty for
modern control
systems.

